

Formalizing Bombieri Vinogradov

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Chapter 1

Introduction

Let Λ be the von Mangoldt function and, for a modulus $q \geq 1$ and a residue class $a \bmod q$, let

$$\psi(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n)$$

be Chebyshev's function in the arithmetic progression $a \bmod q$. The prime number theorem in arithmetic progressions predicts $\psi(x; q, a) \approx x/\varphi(q)$ for $(a, q) = 1$, but the best unconditional estimates for an *individual* modulus q (the Siegel–Walfisz theorem) only cover $q \leq (\log x)^C$, and even the generalised Riemann hypothesis would only give $q \leq x^{1/2-\varepsilon}$. The Bombieri–Vinogradov theorem shows that the latter range is available *on average* over the modulus: for every $A \geq 0$,

$$\sum_{q \leq Q} \max_{y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A} \quad \text{uniformly for } x \geq 2 \text{ and } 1 \leq Q \leq \frac{\sqrt{x}}{(\log x)^{A+3}}.$$

This is often described as “the generalised Riemann hypothesis on average”, and it is the main analytic input in sieve-theoretic applications such as the work of Goldston–Pintz–Yıldırım and Maynard on small gaps between primes.

This blueprint describes a formalisation of the Bombieri–Vinogradov theorem in Lean 4 and Mathlib, following Chapter 26 of Koukoulopoulos’ *The Distribution of Prime Numbers*. Two analytic results are taken as external inputs (axioms in Lean): the Siegel–Walfisz theorem (Assumption 4.1) and the multiplicative large sieve inequality (Assumption 4.6). Everything else is proved from Mathlib and the PrimeNumberTheoremAnd library (from which we borrow the smoothed cutoff function used in the Perron-type argument of Chapter 9).

The text has been written so as to match the Lean development exactly: every numbered statement carries the hypotheses that its Lean counterpart carries, and each proof follows the steps of the formal proof. Implied constants are not tracked: in Lean every $\ll$ below is witnessed by some real number, obtained where convenient by choice or compactness arguments, and nothing is claimed about its size (in particular the implied constants in the Siegel–Walfisz axiom are not effective). Several statements therefore differ from the textbook presentation; the most important deviations are recorded in Chapter 2.

Theorem (Main result, informal form). Assume the Siegel–Walfisz theorem and the large sieve inequality. Then for every $A \in \mathbb{N}$, all real $x \geq 2$ and $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,

$$\sum_{q \leq Q} \sup_{1 \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}.$$

The precise statement is Theorem [10.6](#) (with a real-valued corollary Corollary [10.7](#)).

Chapter 2

Overview of the proof

2.1 Discrepancies and the reduction to Λ

For an arithmetic function f we measure its distribution in progressions by the *discrepancy*

$$\Delta_f(y; q, a) := \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} f(n) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq y \\ (n, q) = 1}} f(n)$$

(Definition 3.7). For $f = \Lambda$ the difference between $\psi(y; q, a) - y/\varphi(q)$ and $\Delta_\Lambda(y; q, a)$ is $\varphi(q)^{-1}(\sum_{n \leq y, (n, q) = 1} \Lambda(n) - y)$, which is controlled by the prime number theorem with logarithmic savings (a special case of Siegel–Walfisz, Theorem 4.2) together with the elementary bound $\sum_{n \leq y, (n, q) > 1} \Lambda(n) \leq \log q \log y / \log 2$ (Lemma 4.3). Summing $1/\varphi(q)$ over $q \leq Q$ costs a factor $\log x$ (Theorem 5.6). The main theorem is thus reduced to the statement Theorem 10.2 about Δ_Λ , in which the maximum over y is restricted to $\sqrt{x} \leq y \leq x$; the range $y < \sqrt{x}$ is handled by the trivial bound $|\psi(y; q, a) - y/\varphi(q)| \ll y$ (Lemma 5.1).

2.2 Vaughan’s identity

Fix parameters U, V with $UV \leq \sqrt{x}$ and $U, V \geq e^{\sqrt{\log x}}$ (Definition 3.13). Vaughan’s identity (Theorem 6.5) decomposes

$$\Lambda = \Lambda^\sharp + \Lambda^\flat + \Lambda_{\leq U}, \quad \Lambda^\sharp = \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1, \quad \Lambda^\flat = (\Lambda_{> U} * 1) * \mu_{> V}.$$

The three pieces are treated separately (Chapters 7 to 9) and recombined by the triangle inequality in Chapter 10. The piece $\Lambda_{\leq U}$ is supported on $[1, U]$ and is bounded trivially. The piece $\Lambda^\sharp$ is a *Type I* sum: a convolution of a coefficient function of small support with the smooth functions $\log$ and 1 , whose discrepancies are bounded pointwise by Abel summation and $|\Delta_1| \leq 1$. The piece $\Lambda^\flat$ is a *Type II* (bilinear) sum, which is bounded on average over q by expanding $\Delta_{\Lambda^\flat}$ in Dirichlet characters, grouping characters by conductor, treating the small conductors with Siegel–Walfisz, and treating the large conductors with the large sieve.

2.3 The two external inputs

Siegel–Walfisz (Assumption 4.1) is used in three places: with $q = 1$ as the prime number theorem (Theorem 4.2), in the reduction from ψ to Δ_Λ , and in the treatment of the small conductors

$d \leq (\log x)^C$ in the Type II sum (Lemma 9.7). The *large sieve* (Assumption 4.6) enters only through the maximal large sieve inequality for convolutions (Theorem 9.37, Theorem 26.6 of Koukoulopoulos), which we prove via a smoothed Perron formula rather than the truncated Perron formula of the textbook (Chapter 9).

2.4 Where the formalisation deviates from the textbook

The following points are worth keeping in mind when comparing with Koukoulopoulos.

- *A divisor factor in the Type I bound.* The pointwise bound for the restricted function $\Lambda_r^\sharp$ carries a factor $\tau(r)$ (Theorem 8.17): restriction to integers coprime to r commutes with convolution (Lemma 8.6), so the smooth factor $\log$ becomes $\log_r$, which is neither smooth nor monotone, and we Möbius-expand it as a sum of $\tau(r)$ dilated copies of $\log$ and 1 (Lemma 8.11). The factor is harmless: in the Type I average it is summed over the modulus using $\sum_{q \leq Q} \tau(q) \leq 3Q \log Q$ (Theorem 5.2), and in the small-conductor analysis it is absorbed by $\tau(q) \leq 2\sqrt{q} \leq 2x^{1/4}$, which is why the small-conductor lemmas assume $q \leq \sqrt{x}$.
- *The large sieve for convolutions is proved by smoothing.* Instead of the truncated Perron formula, we use the smoothed cutoff $\tilde{1}_\varepsilon$ of the PrimeNumberTheoremAnd library (Definition 9.22). At half-integers $y = K + \frac{1}{2}$ the smoothed and sharp cutoffs agree exactly when $\varepsilon = 1/(6 \log 2 \cdot x)$ (Theorem 9.31), so no error term arises, and Mellin inversion on the line $\Re s = 1/\log(x+1)$ (Theorem 9.32) reduces the problem to the large sieve for Dirichlet polynomials (Theorem 9.34) and to a kernel integral of size $O(\log x)$ (Lemma 9.28).
- *Uniformity in U, V .* All Type I/II estimates are proved for an arbitrary parameter datum (Definition 3.13); the choice $U = V = e^{\sqrt{\log x}}$ is only made in the final assembly. The constraints force $\log x \geq 16$ (Lemma 3.14), so the main theorem splits into the range $x \leq e^{32}$, where the trivial bound suffices (Lemma 10.4), and the range $x > e^{32}$.
- *Maxima are extended-real suprema.* The maxima over y and a are formalised as suprema in $[0, \infty]$ (Definition 3.16, Definition 3.17), so that no boundedness hypotheses need to be carried around; finiteness is part of each estimate.

Chapter 3

Definitions and preliminaries

3.1 Notation

$\mathbb{N} = \{0, 1, 2, \dots\}$ denotes the natural numbers (as in Lean), although all sums over $n \leq x$ start at $n = 1$. For $a, b \in \mathbb{N}$ we write $(a, b) = \gcd(a, b)$ and $\tau(n)$ for the number of divisors of n . Dirichlet convolution is written $f * g$, the constant function 1 on $\mathbb{N}_{\geq 1}$ (Mathlib's ζ) is written 1, μ is the Möbius function, φ Euler's totient and Λ the von Mangoldt function; $\log$ also denotes the arithmetic function $n \mapsto \log n$ and $\log^v$ its pointwise v -th power (with $\log^0 = 1$). For a Dirichlet character χ modulo q we write χ_0 for the principal character, $\sum_{\chi \bmod q}^*$ for a sum over primitive characters, and $\xi \uparrow^q$ for the character modulo q induced by a character ξ modulo a divisor of q . Complex conjugation is $\bar{\cdot}$.

For real-valued quantities f, g the statement $f \ll g$ (equivalently $f = O(g)$) means $|f| \leq Cg$ for some constant $C > 0$; a subscript, as in $f \ll_A g$, records the parameters the constant may depend on. Unless stated otherwise the constant may also depend on the implied constants of the two external inputs (Assumption 4.1, Assumption 4.6) and on the fixed bump function of Definition 9.22, but never on x, y, q, a, Q or on the parameter datum. Constants are not tracked in the formalisation.

3.2 Summatory functions and Chebyshev's function

Definition 3.1 (Summatory function). For $f : \mathbb{N} \rightarrow R$ with values in an additive commutative monoid R and a real number x , the *summatory function* of f is

$$\sum_{n \leq x} f(n) := \sum_{1 \leq n \leq \lfloor x \rfloor} f(n).$$

The term $n = 0$ is never included, and the sum is empty (hence 0) when $x < 1$. Throughout the blueprint, every sum of the form $\sum_{n \leq x}$ over a real endpoint x is to be read in this way.

Definition 3.2 (Residue classes). For $q \in \mathbb{N}$ and $a \in \mathbb{Z}/q\mathbb{Z}$, we write $\{n \in \mathbb{N} : n \equiv a \pmod{q}\}$ for the set of natural numbers whose residue class modulo q is a . When $q = 0$ we identify $\mathbb{Z}/0\mathbb{Z} = \mathbb{Z}$, and when $q = 1$ every natural number lies in the single residue class.

Definition 3.3 (Chebyshev's function in arithmetic progressions). For a real number x , a modulus $q \in \mathbb{N}$ and a residue class $a \in \mathbb{Z}/q\mathbb{Z}$, set

$$\psi(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} \Lambda(n),$$

where Λ is the von Mangoldt function. We do not require a to be coprime to q in the definition. For $q = 1$ this is the classical Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$.

3.3 The discrepancy Δ_f

Definition 3.4 (Restriction to integers coprime to r). For $r \in \mathbb{N}$ and $f : \mathbb{N} \rightarrow R$ (with R having a zero), we write f_r for the function

$$f_r(n) := f(n) 1_{(n,r)=1} = \begin{cases} f(n) & \text{if } \gcd(r, n) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Note that f_0 is supported on $n = 1$ only (since $\gcd(0, n) = n$), and $f_1 = f$.

Definition 3.5 (Restriction of an arithmetic function to a set). For an arithmetic function f and a set $S \subseteq \mathbb{N}$, let $f|_S$ denote the arithmetic function $f|_S(n) := f(n) 1_{n \in S}$. In particular $f_r = f|_{\{n : (n,r)=1\}}$ in the notation of Definition 3.4, and $f 1_{[1,U]} = f|_{\{1, \dots, [U]\}}$.

Definition 3.6 (Dilation of an arithmetic function). For $e \geq 1$ and an arithmetic function f , the *dilation* of f by e is the arithmetic function

$$(\delta_e * f)(n) := 1_{e|n} f(n/e),$$

i.e. the Dirichlet convolution of f with the indicator δ_e of the singleton $\{e\}$.

Definition 3.7 (The discrepancy Δ_f). Let R be a field, $f : \mathbb{N} \rightarrow R$, $x \in \mathbb{R}$, $q \in \mathbb{N}$ and $a \in \mathbb{Z}/q\mathbb{Z}$. The *discrepancy* of f in the progression $a \bmod q$ up to x is

$$\Delta_f(x; q, a) := \sum_{\substack{n \leq x \\ n \equiv a \pmod{q}}} f(n) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq x \\ (n,q)=1}} f(n).$$

Here φ is Euler's totient function, and $1/\varphi(0)$ is read as 0 (as is customary in Lean), so that for $q = 0$ only the first sum survives; all estimates below assume $q \geq 1$.

Lemma 3.8 (Linearity of Δ). For fixed x, q, a the map $f \mapsto \Delta_f(x; q, a)$ is R -linear: $\Delta_{f+g} = \Delta_f + \Delta_g$, $\Delta_{f-g} = \Delta_f - \Delta_g$, $\Delta_{cf} = c \Delta_f$ for $c \in R$, and $\Delta_{\sum_{i \in I} f_i} = \sum_{i \in I} \Delta_{f_i}$ for every finite index set I .

Proof. Both sums in Definition 3.7 are finite sums of values of f , hence linear in f ; the finite-sum version follows by induction on I . $\square$

Lemma 3.9 (Trivial bound for Δ_f). For every $f : \mathbb{N} \rightarrow \mathbb{R}$, every real z , every $q \in \mathbb{N}$ and every $a \in \mathbb{Z}/q\mathbb{Z}$,

$$|\Delta_f(z; q, a)| \leq 2 \sum_{n \leq z} |f(n)|.$$

Proof. Each of the two sums in Definition 3.7 is a sub-sum of $\sum_{n \leq z} f(n)$, so each is bounded in absolute value by $\sum_{n \leq z} |f(n)|$; moreover $1/\varphi(q) \leq 1$ (this also holds for $q = 0$, where the factor is 0). The triangle inequality gives the claim. $\square$

Lemma 3.10 (Restricting to integers coprime to the modulus). *Let $f : \mathbb{N} \rightarrow R$ (R a field), $y \in \mathbb{R}$, $q \in \mathbb{N}$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$. Then*

$$\Delta_{f_q}(y; q, a) = \Delta_f(y; q, a),$$

with f_q as in Definition 3.4.

Proof. If $n \equiv a \pmod{q}$ with a a unit, then n is a unit modulo q , i.e. $(n, q) = 1$, so $f_q(n) = f(n)$ on the first sum in Definition 3.7. On the second sum $f_q = f$ by definition. $\square$

Lemma 3.11 (Δ_f via Dirichlet characters). *Let $\mathbb{K}$ be a field equipped with an algebra map $\mathbb{K} \rightarrow \mathbb{C}$ (in practice $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$), $f : \mathbb{N} \rightarrow \mathbb{K}$, $y \in \mathbb{R}$, $q \geq 1$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$. Then, in $\mathbb{C}$,*

$$\Delta_f(y; q, a) = \frac{1}{\varphi(q)} \sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} \overline{\chi(a)} \sum_{n \leq y} f(n) \chi(n),$$

where χ_0 is the principal character modulo q .

Proof. By the orthogonality relation $\sum_{\chi \bmod q} \overline{\chi(a)} \chi(n) = \varphi(q) 1_{n \equiv a \pmod{q}}$, valid for a a unit, summing over all characters gives $\sum_{\chi} \overline{\chi(a)} \sum_{n \leq y} f(n) \chi(n) = \varphi(q) \sum_{n \leq y, n \equiv a \pmod{q}} f(n)$. The principal character contributes $\sum_{n \leq y} f(n) \chi_0(n) = \sum_{n \leq y, (n, q) = 1} f(n)$ since $\chi_0(a) = 1$. Subtracting the principal term and dividing by $\varphi(q)$ gives exactly the two sums in Definition 3.7. (Here $\overline{\chi(a)} = \chi(a)^{-1} = \chi(a^{-1})$ because $|\chi(a)| = 1$ for units a .) $\square$

Lemma 3.12 (Δ_Λ and the Chebyshev error). *For all $z \in \mathbb{R}$, $q \in \mathbb{N}$ and $a \in \mathbb{Z}/q\mathbb{Z}$,*

$$\Delta_\Lambda(z; q, a) = \psi(z; q, a) - \frac{1}{\varphi(q)} \sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n),$$

and consequently

$$\psi(z; q, a) - \frac{z}{\varphi(q)} = \Delta_\Lambda(z; q, a) + \frac{1}{\varphi(q)} \left(\sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n) - z \right).$$

Proof. The first identity is the definition of Δ_Λ together with the definition of $\psi(z; q, a)$ (Definition 3.3); the second follows by adding and subtracting $z/\varphi(q)$. $\square$

3.4 The parameters

Definition 3.13 (The parameters x, U, V). *A parameter datum consists of real numbers x, U, V satisfying*

$$x \geq 2, \quad UV \leq \sqrt{x}, \quad U \geq e^{\sqrt{\log x}}, \quad V \geq e^{\sqrt{\log x}}.$$

Every statement in Chapters 6 to 10 that mentions x, U or V without quantifying them is implicitly universally quantified over such a datum; the constants in those statements never depend on the datum. (In Lean this is the typeclass `ProofData`.) The main theorem is proved by instantiating $U = V = e^{\sqrt{\log x}}$ once $x > e^{32}$, see Theorem 10.6.

Lemma 3.14 (The parameters force $\log x$ to be large). *For every parameter datum (x, U, V) we have $\log x \geq 16$; in particular $\log x \geq 1$, $x \geq e^{16}$, $\sqrt{x} \geq 2$ and $\sqrt{x} \leq x$.*

Proof. Multiplying the two lower bounds for U and V and using $UV \leq \sqrt{x}$ gives $e^{2\sqrt{\log x}} \leq UV \leq \sqrt{x} = e^{(\log x)/2}$, so $2\sqrt{\log x} \leq \frac{1}{2}\log x$. Writing $s = \sqrt{\log x} > 0$ this reads $4s \leq s^2$, i.e. $s \geq 4$, so $\log x \geq 16$. The remaining claims follow: $x \geq e^{16} \geq 4$ gives $\sqrt{x} \geq 2$, and $\sqrt{x} \leq x$ for $x \geq 1$. $\square$

Lemma 3.15 (Range of U and V). *For every parameter datum, $1 \leq U \leq \sqrt{x} \leq x$ and $1 \leq V \leq \sqrt{x} \leq x$; moreover $U, V \geq e$.*

Proof. Since $\sqrt{\log x} \geq 0$ we have $U, V \geq e^{\sqrt{\log x}} \geq 1$, and $\log x \geq 1$ (Lemma 3.14) even gives $U, V \geq e$. Then $U \leq UV \leq \sqrt{x}$ using $V \geq 1$, and symmetrically for V ; finally $\sqrt{x} \leq x$ as $x \geq 1$. $\square$

3.5 Maximal quantities

Definition 3.16 (Maximum over $\sqrt{x} \leq y \leq x$). For a function $F : \mathbb{R} \rightarrow [0, \infty]$ we write

$$\max_{\sqrt{x} \leq y \leq x} F(y) := \sup\{F(y) : \sqrt{x} \leq y \leq x\} \in [0, \infty].$$

The supremum is taken in the extended non-negative reals, so it always exists; all the maxima appearing in the estimates below are of this form (or the analogous Definition 3.17), and it is part of each estimate that the supremum is finite.

Definition 3.17 (Maximum over y and over reduced residues). For $q \in \mathbb{N}$ and $F : \mathbb{R} \times \mathbb{Z}/q\mathbb{Z} \rightarrow [0, \infty]$ we write

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} F(y, a) := \sup_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \sup_{\sqrt{x} \leq y \leq x} F(y, a) \in [0, \infty],$$

the inner supremum being Definition 3.16. The residue a ranges over the units of $\mathbb{Z}/q\mathbb{Z}$ only.

Lemma 3.18 (Basic properties of the maximum). *Let $q \in \mathbb{N}$ and $F : \mathbb{R} \times \mathbb{Z}/q\mathbb{Z} \rightarrow [0, \infty]$.*

1. *If $a \in (\mathbb{Z}/q\mathbb{Z})^*$ and $\sqrt{x} \leq y \leq x$, then $F(y, a) \leq \max_{y,a} F(y, a)$.*
2. *If $F(y, a) \leq M$ for all $a \in (\mathbb{Z}/q\mathbb{Z})^*$ and all $\sqrt{x} \leq y \leq x$, then $\max_{y,a} F(y, a) \leq M$.*

The same holds for Definition 3.16 without the residue variable.

Proof. These are the defining properties of a supremum in the complete lattice $[0, \infty]$. $\square$

Lemma 3.19 (From pointwise bounds to maximal bounds). *Let $f : \mathbb{N} \rightarrow \mathbb{R}$, $q \in \mathbb{N}$ and $B \in \mathbb{R}$. If $|\Delta_f(y; q, a)| \leq B$ for all $\sqrt{x} \leq y \leq x$ and all $a \in (\mathbb{Z}/q\mathbb{Z})^*$, then*

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_f(y; q, a)| \leq B,$$

where the left-hand side is the maximum of Definition 3.17 applied to the (extended) absolute value. Conversely each $|\Delta_f(y; q, a)|$ with a a unit and $\sqrt{x} \leq y \leq x$ is bounded by this maximum.

Proof. Immediate from Lemma 3.18, after identifying the extended-real absolute value $\|\cdot\|_e$ with $|\cdot|$ on $\mathbb{R}$. $\square$

Lemma 3.20 (Triangle inequality for maximal discrepancies). *For all $f, g : \mathbb{N} \rightarrow \mathbb{R}$ and $q \in \mathbb{N}$,*

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{f+g}(y; q, a)| \leq \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_f(y; q, a)| + \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_g(y; q, a)|.$$

Proof. Fix a unit a and $\sqrt{x} \leq y \leq x$. By Lemma 3.8 and the triangle inequality, $|\Delta_{f+g}(y; q, a)| \leq |\Delta_f(y; q, a)| + |\Delta_g(y; q, a)|$, and each term on the right is at most the corresponding maximum by Lemma 3.18. Taking the supremum over y and a on the left (Lemma 3.18 again) gives the claim. $\square$

Chapter 4

External analytic inputs

Two analytic results are quoted from the literature and taken as axioms in Lean. In a proof, “external input” means that the assertion is quoted and no proof is given here. Both are stated with an unspecified constant, which is itself part of the axiom.

4.1 The Siegel–Walfisz theorem

Assumption 4.1 (Siegel–Walfisz). Let $A, C \in \mathbb{N}$. For all real $x \geq 2$, all $1 \leq q \leq (\log x)^C$ and all $a \in (\mathbb{Z}/q\mathbb{Z})^*$,

$$\psi(x; q, a) = \frac{x}{\varphi(q)} + O_{A,C} \left(\frac{x}{(\log x)^A} \right).$$

This is the Siegel–Walfisz theorem in the form “an arbitrary power of log savings, uniformly for $q \leq (\log x)^C$ ”. It is taken as an external input (an axiom in Lean, whose implied constant is an unspecified real number); no proof is given here. Only the cases $C \in \{0, 2C'\}$ are used, see Theorem 4.2 and Lemma 9.7.

Theorem 4.2 (The prime number theorem with logarithmic savings). For every $A \in \mathbb{N}$ and every real $z \geq 2$,

$$\sum_{n \leq z} \Lambda(n) = z + O_A \left(\frac{z}{(\log z)^A} \right).$$

Proof. Apply Assumption 4.1 with $C = 0$, $q = 1$ (note $1 \leq (\log z)^0 = 1$) and $a = 1$: then $\psi(z; 1, 1) = \sum_{n \leq z} \Lambda(n)$ and $\varphi(1) = 1$. $\square$

Lemma 4.3 (Prime powers sharing a factor with q). For every real $z \geq 2$ and every integer $q \geq 1$,

$$0 \leq \sum_{\substack{n \leq z \\ (n, q) > 1}} \Lambda(n) \leq \frac{\log q \cdot \log z}{\log 2}.$$

Proof. The summand vanishes unless $n = p^k$ is a prime power, and $\Lambda(p^k) = \log p$. Grouping by the exponent k , which satisfies $1 \leq k \leq \log z / \log 2$ (as $2^k \leq p^k \leq z$), the sum equals

$$\sum_{1 \leq k \leq \log z / \log 2} \sum_{\substack{p \leq z^{1/k} \\ (p^k, q) > 1}} \log p.$$

For a prime p the condition $(p^k, q) > 1$ means $p \mid q$, so the inner sum is over a subset of the prime divisors of q and is at most $\sum_{p \mid q} \log p \leq \sum_{d \mid q} \Lambda(d) = \log q$. There are at most $\log z / \log 2$ values of k , which gives the bound. Non-negativity is clear. $\square$

Theorem 4.4 (Prime number theorem for integers coprime to q). *For every $B \in \mathbb{N}$, every real $z \geq 2$ and every integer $q \geq 1$,*

$$\sum_{\substack{n \leq z \\ (n, q) = 1}} \Lambda(n) = z + O_B\left(\frac{z}{(\log z)^B}\right) + O(\log q \cdot \log z),$$

the second error term having an absolute implied constant.

Proof. Write $\sum_{n \leq z, (n, q) = 1} \Lambda(n) - z = (\sum_{n \leq z} \Lambda(n) - z) - \sum_{n \leq z, (n, q) > 1} \Lambda(n)$ and apply the triangle inequality: the first term is bounded by Theorem 4.2 and the second by Lemma 4.3. $\square$

Lemma 4.5 (Comparing $z/(\log z)^B$ with $x/(\log x)^B$). *Let $B \in \mathbb{N}$ and let $1 < z \leq x$ be real numbers with $\sqrt{x} \leq z$. Then*

$$\frac{z}{(\log z)^B} \leq 2^B \frac{x}{(\log x)^B}.$$

Proof. From $z \geq \sqrt{x}$ we get $\log z \geq \frac{1}{2} \log x > 0$, so $(\log x)^B \leq 2^B (\log z)^B$. Combining with $z \leq x$ gives $z(\log x)^B \leq 2^B x(\log z)^B$, which is the claim after dividing by the positive quantity $(\log z)^B (\log x)^B$. $\square$

4.2 The large sieve inequality

Assumption 4.6 (The large sieve inequality). For all real $Q \geq 1$, all $H \in \mathbb{Z}$, all integers $N \geq 1$ and all $c : \mathbb{Z} \rightarrow \mathbb{C}$,

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \left| \sum_{H < n \leq H+N} c_n \chi(n) \right|^2 \ll (N + Q^2) \sum_{H < n \leq H+N} |c_n|^2,$$

where $\sum^*$ denotes a sum over primitive characters and the implied constant is absolute. This is the multiplicative large sieve inequality and is taken as an external input (an axiom in Lean, whose implied constant is an unspecified real number); no proof is given here.

Chapter 5

Auxiliary arithmetic estimates

This chapter collects the elementary estimates used in the main argument: the trivial bound for the modular Chebyshev error, the average order of the divisor function, the convergent sums involving $1/\varphi$, and the comparisons between powers of $\log x$ and powers of x .

Lemma 5.1 (Trivial bound for the modular Chebyshev error). *Let $q \geq 1$, let $a \in \mathbb{Z}/q\mathbb{Z}$ and let $z \geq 1$ be real. Then*

$$\left| \psi(z; q, a) - \frac{z}{\varphi(q)} \right| \ll z,$$

with an absolute implied constant.

Proof. Both $\psi(z; q, a)$ and $z/\varphi(q)$ are non-negative, so the absolute value is at most the larger of the two. Since every summand $\Lambda(n)$ is non-negative, $0 \leq \psi(z; q, a) \leq \psi(z)$, and Chebyshev's elementary bound $\psi(z) \ll z$ (Mathlib's `Chebyshev.psi_le_const_mul_self`) applies. On the other hand $\varphi(q) \geq 1$, so $0 \leq z/\varphi(q) \leq z$. $\square$

Theorem 5.2 (Average order of the divisor function). *For every real $Q \geq 2$,*

$$\sum_{q \leq Q} \tau(q) \ll Q \log Q,$$

where $\tau(q)$ is the number of divisors of q .

Proof. Let $N = \lfloor Q \rfloor$. Counting pairs (q, d) with $d \mid q \leq N$ according to d gives the classical identity $\sum_{q \leq N} \tau(q) = \sum_{d \leq N} \lfloor N/d \rfloor$, which is at most $N \sum_{d \leq N} 1/d = NH_N$. The harmonic number satisfies $H_N \leq 1 + \log N \leq 1 + \log Q$, and $1 \ll \log Q$ because $Q \geq 2$. $\square$

Lemma 5.3 (Crude divisor bound). *For every $n \in \mathbb{N}$, $\tau(n) \leq 2\sqrt{n}$.*

Proof. For $n = 0$ both sides vanish (the divisor set of 0 is empty in Lean). For $n \geq 1$ the map $d \mapsto \min(d, n/d)$ sends the divisors of n into $\{1, \dots, \lfloor \sqrt{n} \rfloor\}$, and each value has at most two preimages $\{d, n/d\}$. Hence $\tau(n) \leq 2\lfloor \sqrt{n} \rfloor \leq 2\sqrt{n}$. $\square$

Lemma 5.4 (A lower bound for φ). *For every $d \in \mathbb{N}$, $d \leq 2\varphi(d)^2$.*

Proof. Both sides are multiplicative in d up to the factor 2, so it suffices to check prime powers and to see that the factor 2 is needed at most once. For p^k with $k \geq 1$ one has $\varphi(p^k)^2 = p^{2k-2}(p-1)^2 \geq p^k$ unless $p = 2$, in which case $\varphi(2^k)^2 = 2^{2k-2} \geq 2^k/2$. Multiplying over the prime factorisation of d , in which 2 appears at most once, gives $d \leq 2\varphi(d)^2$; the case $d = 0$ is trivial. $\square$

Lemma 5.5 (A convolution identity for $1/\varphi$). *Let $f_{AF}(d) := \mu(d)^2/(d\varphi(d))$. Then, as arithmetic functions,*

$$\frac{1}{\varphi(n)} = \sum_{d|n} f_{AF}(d) \cdot \frac{1}{e}, \quad \text{i.e.} \quad \frac{1}{\varphi} = f_{AF} * \frac{1}{\text{id}}.$$

Moreover $f_{AF} \geq 0$, $f_{AF}(d) \ll d^{-3/2}$ and $\sum_{d \geq 1} f_{AF}(d)$ converges.

Proof. All three functions $1/\varphi$, f_{AF} and $1/\text{id}$ are multiplicative, so it suffices to compare at prime powers p^k , $k \geq 1$: since $\mu(p^j)^2 = 0$ for $j \geq 2$,

$$(f_{AF} * \frac{1}{\text{id}})(p^k) = \frac{1}{p^k} + \frac{1}{p(p-1)} \cdot \frac{1}{p^{k-1}} = \frac{1}{p^{k-1}} \cdot \frac{p-1+1}{p(p-1)} = \frac{1}{p^{k-1}(p-1)} = \frac{1}{\varphi(p^k)}.$$

For the bound, Lemma 5.4 gives $\varphi(d) \geq \sqrt{d/2}$, whence $f_{AF}(d) \leq 1/(d\varphi(d)) \leq \sqrt{2}d^{-3/2}$, and $\sum d^{-3/2} < \infty$. $\square$

Theorem 5.6 (The sum of $1/\varphi$). *For every parameter datum and every real $Q \leq x$,*

$$\sum_{n \leq Q} \frac{1}{\varphi(n)} \ll \log x,$$

with an absolute implied constant.

Proof. By Lemma 5.5 and the hyperbola identity $\sum_{n \leq Q} (f * g)(n) = \sum_{d \leq Q} f(d) \sum_{e \leq Q/d} g(e)$,

$$\sum_{n \leq Q} \frac{1}{\varphi(n)} = \sum_{d \leq Q} f_{AF}(d) \sum_{e \leq Q/d} \frac{1}{e}.$$

If $Q < 1$ the left side is 0. Otherwise each inner sum is a harmonic number bounded by $1 + \log(Q/d) \leq 1 + \log x \ll \log x$ (as $\log x \geq 1$ by Lemma 3.14), and $\sum_{d \leq Q} f_{AF}(d) \leq \sum_{d \geq 1} f_{AF}(d) < \infty$ since $f_{AF} \geq 0$. $\square$

Lemma 5.7 (The sum of $1/(r\varphi(r))$). *The series $\sum_{d \geq 1} \frac{1}{d\varphi(d)}$ converges, so for every real Q ,*

$$\sum_{r \leq Q} \frac{1}{r\varphi(r)} \ll 1.$$

Proof. By Lemma 5.4, $1/(d\varphi(d)) \leq \sqrt{2}d^{-3/2}$, so the series converges by comparison with $\sum d^{-3/2}$. A partial sum of a series with non-negative terms is at most its sum. $\square$

Lemma 5.8 (Powers of log against powers of x). *For every real $x \geq 1$, every $M \in \mathbb{N}$ and every real $\delta > 0$,*

$$(\log x)^M \ll_{M,\delta} x^\delta.$$

Proof. For $M = 0$ both sides equal 1. For $M \geq 1$ put $c = \delta/M$. Then $\log x = c^{-1} \log(x^c) \leq c^{-1}(x^c - 1) \leq c^{-1}x^c$, using $\log t \leq t - 1$ for $t > 0$. Raising to the M -th power gives $(\log x)^M \leq c^{-M}x^{cM} = (M/\delta)^M x^\delta$. $\square$

Lemma 5.9 (Absorbing powers of $\log x$ into $x/(\log x)^N$). *For every parameter datum, all $j, N \in \mathbb{N}$ and every $q \leq \sqrt{x}$:*

$$(\log x)^j \ll_{j,N} \frac{x}{(\log x)^N}, \quad \sqrt{x} (\log x)^j \ll_{j,N} \frac{x}{(\log x)^N}, \quad \tau(q) \sqrt{x} \log x \ll_N \frac{x}{(\log x)^N}.$$

Proof. By Lemma 5.8 with $M = j + N$ and $\delta = 1/2$, $(\log x)^{j+N} \ll_{j,N} \sqrt{x} \leq x$, which gives the first two bounds after dividing by $(\log x)^N$ (for the second use $\sqrt{x} \cdot \sqrt{x} = x$). For the third, use $\tau(q) \leq 2\sqrt{q} \leq 2x^{1/4}$ (Lemma 5.3) and Lemma 5.8 with $M = N+1$, $\delta = 1/4$: $(\log x)^{N+1} \ll_N x^{1/4}$; then $x^{1/4} \cdot \sqrt{x} \cdot x^{1/4} = x$. $\square$

Lemma 5.10 (Powers of $\log x$ against $\sqrt{U}$). *For every parameter datum, every real $W \geq e^{\sqrt{\log x}}$ and every $n \in \mathbb{N}$,*

$$(\log x)^n \ll_n \sqrt{W}.$$

In particular this applies to $W = U$ and $W = V$.

Proof. Put $s = \sqrt{\log x} \geq 0$, so $(\log x)^n = s^{2n}$. From the Taylor series of the exponential, $(s/2)^{2n}/(2n)! \leq e^{s/2}$, i.e. $s^{2n} \leq 2^{2n}(2n)! e^{s/2}$. Finally $e^{s/2} = \sqrt{e^s} \leq \sqrt{W}$. $\square$

Chapter 6

Vaughan's identity

Throughout the rest of the blueprint a parameter datum (x, U, V) (Definition 3.13) is fixed.

Definition 6.1 (The truncated von Mangoldt function $\Lambda_{\leq U}$). $\Lambda_{\leq U} := \Lambda \cdot 1_{[1, U]}$, i.e. $\Lambda_{\leq U}(n) = \Lambda(n)$ if $n \leq U$ and $\Lambda_{\leq U}(n) = 0$ if $n > U$. We also write $\Lambda_{> U} := \Lambda - \Lambda_{\leq U}$.

Definition 6.2 (The truncated Möbius function $\mu_{\leq V}$). $\mu_{\leq V} := \mu \cdot 1_{[1, V]}$, i.e. $\mu_{\leq V}(n) = \mu(n)$ if $n \leq V$ and $\mu_{\leq V}(n) = 0$ if $n > V$. We also write $\mu_{> V} := \mu - \mu_{\leq V}$.

Definition 6.3 (The Type I part $\Lambda^\sharp$).

$$\Lambda^\sharp := \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1,$$

where $*$ is Dirichlet convolution, $\log$ is the arithmetic function $n \mapsto \log n$ and $1 = \zeta$ is the constant function 1 on $\mathbb{N}_{\geq 1}$.

Definition 6.4 (The Type II part $\Lambda^\flat$).

$$\Lambda^\flat := (\Lambda_{> U} * 1) * \mu_{> V} = ((\Lambda - \Lambda_{\leq U}) * 1) * (\mu - \mu_{\leq V}).$$

Theorem 6.5 (Vaughan's identity). *For every $n \in \mathbb{N}$,*

$$\Lambda(n) = \Lambda^\sharp(n) + \Lambda^\flat(n) + \Lambda_{\leq U}(n).$$

Proof. Expand $\Lambda^\flat$ by bilinearity of the Dirichlet convolution and use the identities $\Lambda * 1 = \log$ and $1 * \mu = \delta_1$ (the convolution identity):

$$\Lambda^\flat = \Lambda * 1 * \mu - \Lambda * 1 * \mu_{\leq V} - \Lambda_{\leq U} * 1 * \mu + \Lambda_{\leq U} * 1 * \mu_{\leq V} = \Lambda - \log * \mu_{\leq V} - \Lambda_{\leq U} + \Lambda_{\leq U} * \mu_{\leq V} * 1.$$

Adding $\Lambda^\sharp = \mu_{\leq V} * \log - \Lambda_{\leq U} * \mu_{\leq V} * 1$ and $\Lambda_{\leq U}$, everything cancels except Λ . $\square$

Chapter 7

The small part $\Lambda_{\leq U}$

The contribution of $\Lambda_{\leq U}$ is bounded trivially: it is supported on $[1, U]$ with $U \leq \sqrt{x}$, and there are at most $Q \leq \sqrt{x}/(\log x)^{A+3}$ moduli.

Lemma 7.1 (The ℓ^1 norm of $\Lambda_{\leq U}$). *For every parameter datum and every real y ,*

$$\sum_{n \leq y} \Lambda_{\leq U}(n) \leq U \log x.$$

Proof. The function $\Lambda_{\leq U}$ is non-negative, supported on the at most U integers $1 \leq n \leq U$, and on its support $\Lambda_{\leq U}(n) = \Lambda(n) \leq \log n \leq \log U \leq \log x$ by Lemma 3.15. Hence the sum over any range is at most $U \log x$. $\square$

Lemma 7.2 (Pointwise bound for $\Delta_{\Lambda_{\leq U}}$). *For every parameter datum, every real y , every $q \geq 1$ and every $a \in \mathbb{Z}/q\mathbb{Z}$,*

$$|\Delta_{\Lambda_{\leq U}}(y; q, a)| \leq 2U \log x.$$

Proof. Both sums in the definition of $\Delta_{\Lambda_{\leq U}}(y; q, a)$ are sums of non-negative terms bounded by $\sum_{n \leq y} \Lambda_{\leq U}(n) \leq U \log x$ (Lemma 7.1), and $1/\varphi(q) \leq 1$. Hence the difference has absolute value at most $2U \log x$. $\square$

Theorem 7.3 (The small part on average). *For every parameter datum, every $A \in \mathbb{N}$ and every real $0 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda_{\leq U}}(y; q, a)| \leq \frac{2x}{(\log x)^{A+2}}.$$

Proof. By Lemma 7.2 and Lemma 3.19, each of the $\lfloor Q \rfloor \leq Q$ summands is at most $2U \log x$. Using $U \leq \sqrt{x}$ (Lemma 3.15) and the hypothesis on Q ,

$$Q \cdot 2U \log x \leq \frac{\sqrt{x}}{(\log x)^{A+3}} \cdot 2\sqrt{x} \log x = \frac{2x}{(\log x)^{A+2}}.$$

$\square$

Chapter 8

Type I sums: $\Lambda^\sharp$

8.1 Discrepancy of smooth weights

The basic Type I mechanism: for f of small support, Δ_{f*g} is a short sum of discrepancies of g at the points y/k (Lemma 8.1); when $g = \log^v$ these are bounded pointwise via Abel summation against the trivial bound $|\Delta_1| \leq 1$.

Lemma 8.1 (Δ of a Dirichlet convolution). *Let $\mathbb{K}$ be a field with an algebra map to $\mathbb{C}$, let f, g be arithmetic functions with values in $\mathbb{K}$, let $y \in \mathbb{R}$, $q \geq 1$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$. Then*

$$\Delta_{f*g}(y; q, a) = \sum_{\substack{k \leq y \\ (k, q) = 1}} f(k) \Delta_g\left(\frac{y}{k}; q, a\bar{k}\right),$$

where $\bar{k}$ denotes the inverse of k in $(\mathbb{Z}/q\mathbb{Z})^*$.

Proof. Both sides may be computed in $\mathbb{C}$ after applying the algebra map. By Lemma 3.11, $\varphi(q)\Delta_{f*g}(y; q, a) = \sum_{\chi \neq \chi_0} \overline{\chi(a)} \sum_{n \leq y} (f * g)(n) \chi(n)$. Since χ is completely multiplicative, $(f * g)(n) \chi(n) = ((f\chi) * (g\chi))(n)$, and the hyperbola identity $\sum_{n \leq y} (F * G)(n) = \sum_{k \leq y} F(k) \sum_{l \leq y/k} G(l)$ gives

$$\varphi(q)\Delta_{f*g}(y; q, a) = \sum_{k \leq y} f(k) \sum_{\chi \neq \chi_0} \overline{\chi(a)} \chi(k) \sum_{l \leq y/k} g(l) \chi(l).$$

If $(k, q) > 1$ then $\chi(k) = 0$ and the term vanishes. If $(k, q) = 1$ then k is a unit and $\overline{\chi(a)} \chi(k) = \overline{\chi(a\bar{k})}$, so by Lemma 3.11 (applied to g at the point y/k and the unit $a\bar{k}$) the inner double sum equals $\varphi(q)\Delta_g(y/k; q, a\bar{k})$. Dividing by $\varphi(q)$ gives the claim. $\square$

Lemma 8.2 (The discrepancy of the constant function 1). *For every real x , every $q \geq 1$ and every $a \in \mathbb{Z}/q\mathbb{Z}$ (not necessarily a unit),*

$$|\Delta_1(x; q, a)| \leq 1.$$

Proof. Let $N = \lfloor x \rfloor$ and write $N = kq + m$ with $0 \leq m < q$. The integers $1 \leq n \leq N$ form k complete residue systems modulo q followed by m further integers. Each complete system contains exactly one $n \equiv a \pmod{q}$ and exactly $\varphi(q)$ integers coprime to q . Hence

$$\#\{n \leq N : n \equiv a\} = k + c, \quad \#\{n \leq N : (n, q) = 1\} = \varphi(q)k + c',$$

with $0 \leq c \leq 1$ and $0 \leq c' \leq \varphi(q)$ counting the contributions of the last m integers. Therefore $\Delta_1(x; q, a) = c - c'/\varphi(q) \in [-1, 1]$. $\square$

Lemma 8.3 (Abel summation for Δ). *Let $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$ and $x \geq 1$. Let $g, g' : \mathbb{R} \rightarrow \mathbb{C}$ be such that g has derivative $g'(t)$ at every $t \in [1, x]$, g' is locally integrable on $[1, x]$, and $g(1) = 0$. Then*

$$\Delta_g(x; q, a) = \Delta_1(x; q, a) g(x) - \int_1^x \Delta_1(t; q, a) g'(t) dt,$$

where Δ_g is the discrepancy of the arithmetic function $n \mapsto g(n)$.

Proof. By Lemma 3.11 both sides are, up to the factor $1/\varphi(q)$, sums over the non-principal characters $\chi \bmod q$ of $\chi(a)$ times a character sum; it suffices to prove the identity character by character, and the integral may be exchanged with the finite sum because each integrand is the product of the locally integrable g' with the locally bounded step function $t \mapsto \sum_{n \leq t} \chi(n)$. For a fixed χ put $S(t) = \sum_{n \leq t} \chi(n)$; Abel summation (Mathlib's `sum_mul_eq_sub_sub_integral_mul`) gives

$$\sum_{1 < n \leq x} g(n) \chi(n) = g(x) S(x) - g(1) S(1) - \int_1^x g'(t) S(t) dt,$$

and the term $n = 1$ as well as the boundary term at 1 vanish because $g(1) = 0$. Recombining over χ gives the identity with Δ_1 in place of S . $\square$

Lemma 8.4 (Discrepancy of a monotone weight). *Let $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$ and $x \geq 1$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at every point of $[1, x]$, with g' locally integrable on $[1, x]$, monotone increasing on $[1, x]$ and $g(1) = 0$. Then*

$$|\Delta_g(x; q, a)| \leq 2g(x).$$

Proof. Apply Lemma 8.3 (to the real functions g, g' , viewed in $\mathbb{C}$) and bound the two terms. By Lemma 8.2, $|\Delta_1(x; q, a) g(x)| \leq g(x)$, where $g(x) \geq g(1) = 0$ by monotonicity. Since g is monotone on $[1, x]$, its derivative satisfies $g' \geq 0$ on $(1, x)$, so

$$\left| \int_1^x \Delta_1(t; q, a) g'(t) dt \right| \leq \int_1^x g'(t) dt = g(x) - g(1) = g(x)$$

by the fundamental theorem of calculus. Adding the two bounds gives $2g(x)$. $\square$

Theorem 8.5 (Type I estimate: Δ of $f * \log^v$). *Let $v \in \mathbb{N}$ and let f be a real arithmetic function. For every real $x \geq 2$, every $q \geq 1$ and every $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$|\Delta_{f * \log^v}(x; q, a)| \leq 2(\log x)^v \sum_{k \leq x} |f(k)|,$$

where $\log^v$ denotes the arithmetic function $n \mapsto (\log n)^v$ (so $\log^0 = 1$).

Proof. By Lemma 8.1, $\Delta_{f * \log^v}(x; q, a) = \sum_{k \leq x, (k, q) = 1} f(k) \Delta_{\log^v}(x/k; q, a\bar{k})$, so by the triangle inequality it suffices to show $|\Delta_{\log^v}(x/k; q, b)| \leq 2(\log x)^v$ for every unit b and every $1 \leq k \leq x$. If $v = 0$ this is Lemma 8.2. If $v \geq 1$, apply Lemma 8.4 with $g(t) = (\log t)^v$ on $[1, x/k]$ (note $x/k \geq 1$): g is differentiable with locally integrable derivative $v(\log t)^{v-1}/t$, monotone on $[1, x/k]$ and $g(1) = 0$, so $|\Delta_{\log^v}(x/k; q, b)| \leq 2(\log(x/k))^v \leq 2(\log x)^v$. $\square$

8.2 Restriction to coprime integers and Möbius expansion

We need bounds not for $\Lambda^\sharp$ itself but for its restriction $\Lambda_r^\sharp$ to integers coprime to r (this is what the conductor analysis of Chapter 9 produces). Restriction is a convolution homomorphism, so the smooth factor log becomes the restricted function $\log_r$, to which Theorem 8.5 does not apply directly. We rewrite $\log_r$ (and 1_r) as a Möbius sum of dilations of the unrestricted functions; this costs a factor $\tau(r)$.

Lemma 8.6 (Restriction is a convolution homomorphism). *Let $S \subseteq \mathbb{N}$ be multiplicatively saturated: $ab \in S$ if and only if $a \in S$ and $b \in S$. Then for all arithmetic functions f, g (with values in a semiring),*

$$(f * g)|_S = f|_S * g|_S.$$

*In particular, since $(ab, r) = 1$ if and only if $(a, r) = 1$ and $(b, r) = 1$, this applies to $S = \{n : (n, r) = 1\}$: $(f * g)_r = f_r * g_r$ for every $r \in \mathbb{N}$.*

Proof. Evaluate both sides at n : the left side is $1_{n \in S} \sum_{ab=n} f(a)g(b)$ and the right side is $\sum_{ab=n} 1_{a \in S} 1_{b \in S} f(a)g(b)$. For each pair $ab = n$ saturation gives $1_{n \in S} = 1_{a \in S} 1_{b \in S}$. $\square$

Lemma 8.7 (Dilation commutes with convolution). *For $e \geq 1$ and real arithmetic functions f, g ,*

$$f * (\delta_e * g) = \delta_e * (f * g) \quad \text{and} \quad (\delta_e * f) * g = f * (\delta_e * g).$$

Proof. Both are instances of associativity and commutativity of Dirichlet convolution, with $\delta_e * g$ the dilation of Definition 3.6. Concretely, at $n = em$ both sides of the first identity equal $\sum_{ab=m} f(a)g(b)$ after the substitution $b \mapsto eb$ in the divisor sum, and both vanish when $e \nmid n$. $\square$

Lemma 8.8 (Dilation does not increase the ℓ^1 norm). *For $e \geq 1$, every real arithmetic function f and every real x ,*

$$\sum_{k \leq x} |(\delta_e * f)(k)| \leq \sum_{k \leq x} |f(k)|.$$

Proof. The non-zero terms on the left have $k = em$ with $m \leq x/e$, and $|(\delta_e * f)(em)| = |f(m)|$; reindexing by m gives $\sum_{m \leq x/e} |f(m)|$, a sub-sum of the right-hand side since $x/e \leq x$ (the terms being non-negative). $\square$

Lemma 8.9 (Submultiplicativity of the ℓ^1 norm). *For real arithmetic functions f, g and every real $x \geq 0$,*

$$\sum_{k \leq x} |(f * g)(k)| \leq \left(\sum_{k \leq x} |f(k)| \right) \left(\sum_{k \leq x} |g(k)| \right).$$

Proof. By the triangle inequality the left side is at most $\sum_{k \leq x} \sum_{ab=k} |f(a)||g(b)|$. The map $(a, b) \mapsto (a, b)$ from pairs with $ab = k \leq x$ into $\{1 \leq a \leq x\} \times \{1 \leq b \leq x\}$ is injective (with disjoint images for distinct k), so this double sum is at most $\sum_{a \leq x} \sum_{b \leq x} |f(a)||g(b)|$. $\square$

Lemma 8.10 (Möbius inversion of the coprimality indicator). *For every $r \geq 1$ and every $b \in \mathbb{N}$,*

$$\sum_{e|r} \mu(e) 1_{e|b} = 1_{(r,b)=1}.$$

Proof. The divisors e of r with $e \mid b$ are exactly the divisors of $\gcd(r, b)$, and $\sum_{e|m} \mu(e) = 1_{m=1}$ for $m \geq 1$ (this is $\mu * 1 = \delta_1$). Note that $r \geq 1$ is needed: for $r = 0$ the left side is an empty sum while $(0, 1) = 1$. $\square$

Lemma 8.11 (Möbius expansion of 1_r and $\log_r$). *Let $r \geq 1$. As arithmetic functions,*

$$1_r = \sum_{e|r} \mu(e) (\delta_e * 1), \quad \log_r = \sum_{e|r} \mu(e) (\log e \cdot (\delta_e * 1) + \delta_e * \log),$$

where 1_r and $\log_r$ are the restrictions (Definition 3.4) of the constant function 1 and of $\log$ to the integers coprime to r , and $\delta_e * (\cdot)$ is the dilation of Definition 3.6. (The identity for $\log$ holds for $r = 0$ as well, both sides then vanishing except at $b = 1$ where $\log 1 = 0$.)

Proof. Evaluate at $b \geq 1$. Since $(\delta_e * 1)(b) = 1_{e|b}$, the first identity is Lemma 8.10. For the second, $(\delta_e * \log)(b) = 1_{e|b} \log(b/e)$, so $\log e \cdot (\delta_e * 1)(b) + (\delta_e * \log)(b) = 1_{e|b} (\log e + \log(b/e)) = 1_{e|b} \log b$; summing against $\mu(e)$ over $e | r$ and using Lemma 8.10 gives $1_{(r,b)=1} \log b = \log_r(b)$. At $b = 0$ all terms vanish. $\square$

Lemma 8.12 (Convolution with a restricted analytic factor). *Let H be a real arithmetic function and $r \geq 1$. Then, as functions on $\mathbb{N}$,*

$$H * 1_r = \sum_{e|r} \mu(e) ((\delta_e * H) * 1), \quad H * \log_r = \sum_{e|r} \mu(e) (\log e \cdot ((\delta_e * H) * 1) + (\delta_e * H) * \log).$$

(The second identity holds for all $r \in \mathbb{N}$.)

Proof. Convolve the identities of Lemma 8.11 with H , using bilinearity of the convolution and $H * (\delta_e * g) = (\delta_e * H) * g$ (Lemma 8.7). $\square$

Lemma 8.13 (Type I estimate for a dilated coefficient function). *Let $v \in \mathbb{N}$, $e \geq 1$, let h be a real arithmetic function, $x \geq 2$, $q \geq 1$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$. Then*

$$|\Delta_{(\delta_e * h) * \log^v}(x; q, a)| \leq 2(\log x)^v \sum_{k \leq x} |h(k)|.$$

Proof. Apply Theorem 8.5 to the arithmetic function $\delta_e * h$ and then Lemma 8.8. $\square$

8.3 The pointwise bound for $\Lambda_r^\sharp$

Lemma 8.14 (ℓ^1 norms of the coefficient functions). *For every parameter datum and every real x ,*

$$\sum_{k \leq x} |\mu_{\leq V}(k)| \leq V, \quad \text{and, if } x \geq 2, \quad \sum_{k \leq x} |\Lambda_{\leq U}(k)| \leq U \log x.$$

Proof. The function $\mu_{\leq V}$ is supported on the at most V integers $1 \leq k \leq V$ and satisfies $|\mu_{\leq V}| \leq 1$. The function $\Lambda_{\leq U}$ is non-negative, supported on $1 \leq k \leq U$, and $\Lambda(k) \leq \log k$. If $U \leq x$ its support has at most U points each contributing at most $\log U \leq \log x$; if $x < U$ the range $k \leq x$ has at most $x \leq U$ points each contributing at most $\log x$. $\square$

Lemma 8.15 (The first term of $\Lambda_r^\sharp$). *For every parameter datum, every $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$, every $r \in \mathbb{N}$ with $r \leq x$ and every real $2 \leq y \leq x$,*

$$|\Delta_{(\mu_{\leq V} * \log)_r}(y; q, a)| \leq 4\tau(r) V \log x.$$

Proof. By Lemma 8.6, $(\mu_{\leq V} * \log)_r = (\mu_{\leq V})_r * \log_r$, and by Lemma 8.12 with $H = (\mu_{\leq V})_r$,

$$(\mu_{\leq V} * \log)_r = \sum_{e|r} \mu(e) \left(\log e \cdot (\delta_e * H) * 1 + (\delta_e * H) * \log \right).$$

(If $r = 0$ both sides are supported on $n = 1$, where $\log 1 = 0$, so the identity is trivial.) By linearity (Lemma 3.8) and the triangle inequality it suffices to bound each of the $\tau(r)$ summands by $4V \log x$. Fix $e \mid r$; then $1 \leq e \leq r \leq x$, so $0 \leq \log e \leq \log x$, and $|\mu(e)| \leq 1$. By Lemma 8.13 with $v = 0$ and $v = 1$ at the point $y \geq 2$,

$$|\Delta_{(\delta_e * H) * 1}(y; q, a)| \leq 2\|H\|_1, \quad |\Delta_{(\delta_e * H) * \log}(y; q, a)| \leq 2 \log y \|H\|_1,$$

where $\|H\|_1 = \sum_{k \leq y} |H(k)| \leq \sum_{k \leq y} |\mu_{\leq V}(k)| \leq V$ by Lemma 8.14. Since $\log y \leq \log x$, the e -th summand is at most $\log x \cdot 2V + 2 \log x \cdot V = 4V \log x$. $\square$

Lemma 8.16 (The second term of $\Lambda_r^\sharp$). *For every parameter datum, every $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$, every $r \in \mathbb{N}$ with $r \leq x$ and every real $2 \leq y \leq x$,*

$$|\Delta_{(\Lambda_{\leq U} * \mu_{\leq V} * 1)_r}(y; q, a)| \leq 2\tau(r)UV \log x.$$

Proof. If $r = 0$, the restricted function is supported on $n = 1$ where $(\Lambda_{\leq U} * \mu_{\leq V} * 1)(1) = \Lambda(1)\mu(1) = 0$, so the discrepancy vanishes. Let $r \geq 1$. By Lemma 8.6 and Lemma 8.12 with $H = (\Lambda_{\leq U} * \mu_{\leq V})_r$,

$$(\Lambda_{\leq U} * \mu_{\leq V} * 1)_r = H * 1_r = \sum_{e|r} \mu(e) (\delta_e * H) * 1.$$

By Lemma 3.8, the triangle inequality, $|\mu(e)| \leq 1$ and Lemma 8.13 with $v = 0$, each of the $\tau(r)$ summands has discrepancy at most $2\|H\|_1$, where

$$\|H\|_1 = \sum_{k \leq y} |H(k)| \leq \sum_{k \leq y} |(\Lambda_{\leq U} * \mu_{\leq V})(k)| \leq \left(\sum_{k \leq y} |\Lambda_{\leq U}(k)| \right) \left(\sum_{k \leq y} |\mu_{\leq V}(k)| \right) \leq U \log y \cdot V$$

by Lemma 8.9 and Lemma 8.14. Since $\log y \leq \log x$ the claim follows. $\square$

Theorem 8.17 (Pointwise Type I bound for $\Lambda_r^\sharp$). *For every parameter datum, every $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$, every $r \in \mathbb{N}$ with $r \leq x$ and every real $2 \leq y \leq x$,*

$$|\Delta_{\Lambda_r^\sharp}(y; q, a)| \ll \tau(r)UV \log x,$$

with an absolute implied constant. The factor $\tau(r)$ cannot be dropped: the discrepancy of $\log_r$ is a signed combination of $2^{\omega(r)}$ residue-class counts with no cancellation in general, and the Möbius expansion above records this honestly. Downstream it is harmless because τ is only ever averaged over the modulus (Theorem 8.18) or bounded by $2x^{1/4}$ (Theorem 9.8).

Proof. Since restriction to integers coprime to r is linear, $\Lambda_r^\sharp = (\mu_{\leq V} * \log)_r - (\Lambda_{\leq U} * \mu_{\leq V} * 1)_r$, so by Lemma 3.8 and the triangle inequality $|\Delta_{\Lambda_r^\sharp}(y; q, a)| \leq 4\tau(r)V \log x + 2\tau(r)UV \log x$ by Lemma 8.15 and Lemma 8.16. As $U \geq 1$ (Lemma 3.15), this is at most $6\tau(r)UV \log x$. $\square$

8.4 Summing over the modulus

Theorem 8.18 (The Type I part on average). *For every parameter datum, every $A \in \mathbb{N}$ and every real $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^\sharp}(y; q, a)| \ll \frac{x}{(\log x)^A},$$

with an absolute implied constant.

Proof. Since $\log x \geq 1$, the hypothesis gives $Q \leq \sqrt{x} \leq x$; also $\sqrt{x} \geq 2$ (Lemma 3.14), so every $y \in [\sqrt{x}, x]$ satisfies $2 \leq y \leq x$.

Pointwise bound. Fix $1 \leq q \leq Q$, a unit a and $\sqrt{x} \leq y \leq x$. By Lemma 3.10, $\Delta_{\Lambda^\sharp}(y; q, a) = \Delta_{\Lambda_q^\sharp}(y; q, a)$, and Theorem 8.17 with $r = q \leq x$ gives $|\Delta_{\Lambda^\sharp}(y; q, a)| \ll \tau(q) UV \log x$. By Lemma 3.19 the same bound holds for the maximum over y and a .

Averaging the divisor function. Summing over $q \leq Q$,

$$\sum_{q \leq Q} \max_{y, a} |\Delta_{\Lambda^\sharp}(y; q, a)| \ll UV \log x \sum_{q \leq Q} \tau(q) \ll UV \log x \cdot Q \log x,$$

where for $Q \geq 2$ we used Theorem 5.2 and $\log Q \leq \log x$, while for $1 \leq Q < 2$ the sum is $\tau(1) = 1 \leq Q \log x$.

Conclusion. Using $UV \leq \sqrt{x}$ and $Q(\log x)^{A+2} \leq Q(\log x)^{A+3} \leq \sqrt{x}$,

$$UV \cdot Q(\log x)^2 \cdot (\log x)^A = UV \cdot Q(\log x)^{A+2} \leq \sqrt{x} \cdot \sqrt{x} = x,$$

so the right-hand side above is $\ll x/(\log x)^A$. □

Chapter 9

Type II sums: Λ^b

9.1 Definitions

Definition 9.1 (The twisted sums $S_r(y, \xi)$). For $r \in \mathbb{N}$, a real y and a Dirichlet character ξ modulo some $d \in \mathbb{N}$, set

$$S_r(y, \xi) := \left| \sum_{n \leq y} \Lambda_r^b(n) \xi(n) \right|,$$

where Λ_r^b is the restriction of Λ^b to the integers coprime to r (Definition 3.4).

Definition 9.2 (The quantities $T_r(Q)$). For a real C , an integer $r \geq 1$ and a real Q , set

$$T_r^{(C)}(Q) := \sum_{(\log x)^C < d \leq Q/r} \frac{1}{\varphi(d)} \sum_{\xi \bmod d}^* \max_{\sqrt{x} \leq y \leq x} S_r(y, \xi) \in [0, \infty],$$

where $\sum^*$ is a sum over primitive characters and the maximum is that of Definition 3.16. (For $r = 0$ the range of d is empty and $T_r^{(C)}(Q) = 0$.) We usually suppress C from the notation.

9.2 Characters and conductors

By orthogonality, $\varphi(q)\Delta_{\Lambda^b}(y; q, a)$ is a sum over non-principal characters $\chi \bmod q$. Grouping the characters by conductor $d \mid q$ produces, for each d , a sum over the primitive characters modulo d of the twisted sums $S_{q/d}(y, \xi)$. The conductors $d \leq (\log x)^C$ are treated in Section 9.3 by Siegel–Walfisz, after converting the primitive-character sums back into discrepancies by Möbius inversion; the conductors $d > (\log x)^C$ are collected into the quantities T_r and treated by the large sieve.

Lemma 9.3 (Grouping characters by conductor). *Let $q \geq 1$ and let f be a function on the Dirichlet characters modulo q with values in an additive commutative monoid. Then*

$$\sum_{\substack{\chi \bmod q \\ \chi \neq \chi_0}} f(\chi) = \sum_{\substack{d \mid q \\ d \neq 1}} \sum_{\xi \bmod d}^* f(\xi \uparrow^q),$$

where $\xi \uparrow^q$ denotes the character modulo q induced by ξ , i.e. $(\xi \uparrow^q)(n) = \xi(n)1_{(n, q)=1}$.

Proof. Group the characters $\chi \bmod q$ according to their conductor d , which divides q . For a fixed $d \mid q$, the map $\xi \mapsto \xi \uparrow^q$ is a bijection from the primitive characters modulo d onto the characters modulo q of conductor d : it is injective, it preserves the conductor, and every character of conductor d is induced by its (primitive) associated character modulo d . Finally a character modulo q is principal if and only if its conductor is 1, which accounts for the exclusion of $d = 1$ on the right and of χ_0 on the left. $\square$

Lemma 9.4 ($\varphi(d)\Delta_g$ as a sum over conductors). *Let $q \geq 1$, $a \in (\mathbb{Z}/q\mathbb{Z})^*$, and let $g : \mathbb{N} \rightarrow \mathbb{R}$ vanish at every n with $(n, q) > 1$. For $x \in \mathbb{R}$ and $i \geq 1$ put*

$$G_i := \sum_{\xi \bmod i}^* \overline{\xi(a)} \sum_{n \leq x} \xi(n) g(n).$$

Then for every divisor d of q ,

$$\varphi(d) \Delta_g(x; d, a) = \sum_{\substack{i \mid d \\ i \neq 1}} G_i,$$

where a is reduced modulo d (resp. modulo i) on the right.

Proof. Since $d \mid q$, the reduction of a modulo d is a unit, so Lemma 3.11 gives $\varphi(d)\Delta_g(x; d, a) = \sum_{\chi \neq \chi_0 \bmod d} \overline{\chi(a)} \sum_{n \leq x} \chi(n) g(n)$. Apply Lemma 9.3 to the modulus d . For ξ primitive modulo $i \mid d$ and n with $g(n) \neq 0$ we have $(n, q) = 1$, hence $(n, d) = 1$ and $(\xi \uparrow^d)(n) = \xi(n)$; likewise $(\xi \uparrow^d)(a) = \xi(a)$ because a is a unit. So the summand attached to $\xi \uparrow^d$ is exactly the summand of G_i . $\square$

Lemma 9.5 (Primitive character sums via Möbius inversion). *Let f be a real arithmetic function, $r \in \mathbb{N}$, $q > 1$, $x \in \mathbb{R}$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$. Then*

$$\sum_{\xi \bmod q}^* \overline{\xi(a)} \sum_{n \leq x} \xi(n) f_r(n) = \sum_{d \mid q} \mu(q/d) \varphi(d) \Delta_{f_{rq}}(x; d, a).$$

Proof. Put $g := f_{rq}$; then $g(n) = 0$ whenever $(n, q) > 1$. Since $\xi(n) = 0$ unless $(n, q) = 1$, and $f_r(n) = f_{rq}(n)$ when $(n, q) = 1$, the left-hand side equals G_q in the notation of Lemma 9.4. Define $G'_i := G_i$ for $i \mid q$, $i \neq 1$, and $G'_1 := 0$, and set $F(d) := \sum_{i \mid d} G'_i$ for $d \mid q$. By Lemma 9.4, $F(d) = \varphi(d)\Delta_g(x; d, a)$. Möbius inversion of the relation $F = G' * 1$ on the divisors of q gives $G'_q = \sum_{d \mid q} \mu(q/d) F(d)$, and $G'_q = G_q$ because $q \neq 1$. $\square$

Theorem 9.6 (Decomposing Δ_{Λ^b} by conductor). *For every parameter datum, every $C \in \mathbb{N}$, every real y , every $q \geq 1$ and every $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$|\Delta_{\Lambda^b}(y; q, a)| \leq \frac{1}{\varphi(q)} \left| \sum_{\substack{d \mid q \\ 1 < d \leq (\log x)^C}} \sum_{s \mid d} \mu(d/s) \varphi(s) \Delta_{\Lambda_q^b}(y; s, a) \right| + \frac{1}{\varphi(q)} \sum_{\substack{d \mid q \\ d > (\log x)^C}} \sum_{\xi \bmod d}^* S_{q/d}(y, \xi).$$

Proof. By Lemma 3.10, $\Delta_{\Lambda^b}(y; q, a) = \Delta_{\Lambda_q^b}(y; q, a)$, and Lemma 9.4 with $g = \Lambda_q^b$ and $d = q$ gives

$$\varphi(q) \Delta_{\Lambda^b}(y; q, a) = \sum_{\substack{d \mid q \\ d \neq 1}} G_d, \quad G_d = \sum_{\xi \bmod d}^* \overline{\xi(a)} \sum_{n \leq y} \xi(n) \Lambda_q^b(n).$$

Split the divisors $d \neq 1$ of q according to whether $d \leq (\log x)^C$ or $d > (\log x)^C$ and apply the triangle inequality to the two groups.

For $1 < d \leq (\log x)^C$, Lemma 9.5 (with modulus d , $r = q$ and $f = \Lambda^b$) expresses G_d as $\sum_{s|d} \mu(d/s) \varphi(s) \Delta_{\Lambda_{qd}^b}(y; s, a)$, and $\Lambda_{qd}^b = \Lambda_q^b$ since $d \mid q$. This is the first term.

For $d > (\log x)^C$ we bound $|G_d|$ termwise: $|\xi(a)| = 1$, and for a character ξ modulo d one has $\xi(n) \Lambda_q^b(n) = \xi(n) \Lambda_{q/d}^b(n)$ for all n , because both sides vanish unless $(n, d) = 1$, in which case $(n, q) = 1 \iff (n, q/d) = 1$. Hence $|G_d| \leq \sum_{\xi}^* S_{q/d}(y, \xi)$, which is the second term. $\square$

9.3 Small conductors

Lemma 9.7 (Siegel–Walfisz for Λ_q). *Let $A', C_2 \in \mathbb{N}$. For every parameter datum, every real $2 \leq y \leq x$, all integers $1 \leq s \mid q$ with $q \leq \sqrt{x}$ and $s \leq (\log y)^{C_2}$, and every $a \in (\mathbb{Z}/s\mathbb{Z})^*$,*

$$|\Delta_{\Lambda_q}(y; s, a)| \ll_{A', C_2} \frac{y}{(\log y)^{A'}} + (\log x)^2.$$

Proof. Write $\Lambda_q = \Lambda - \Lambda^{\text{nc}}$ where $\Lambda^{\text{nc}}(n) = \Lambda(n) 1_{(n, q) > 1}$, and use Lemma 3.8.

The non-coprime part. Since $\Lambda^{\text{nc}} \geq 0$, Lemma 3.9 and Lemma 4.3 give $|\Delta_{\Lambda^{\text{nc}}}(y; s, a)| \leq 2 \sum_{n \leq y, (n, q) > 1} \Lambda(n) \ll \log q \log y \leq (\log x)^2$, using $q \leq x$ and $y \leq x$.

The main part. By Lemma 3.12,

$$\Delta_{\Lambda}(y; s, a) = \left(\psi(y; s, a) - \frac{y}{\varphi(s)} \right) + \frac{1}{\varphi(s)} \left(y - \sum_{\substack{n \leq y \\ (n, s) = 1}} \Lambda(n) \right).$$

The first bracket is $\ll_{A', C_2} y/(\log y)^{A'}$ by Assumption 4.1, since $1 \leq s \leq (\log y)^{C_2}$ and $y \geq 2$.

The second is bounded, using $1/\varphi(s) \leq 1$ and Theorem 4.4 with $B = A'$, by $\ll_{A'} y/(\log y)^{A'} + \log s \log y$, and $\log s \log y \leq (\log x)^2$. Adding the three contributions gives the claim. $\square$

Theorem 9.8 (Siegel–Walfisz for Λ_q^b). *Let $A, C \in \mathbb{N}$. For every parameter datum, every real $\sqrt{x} \leq y \leq x$, all integers $1 \leq s \mid q$ with $q \leq \sqrt{x}$ and $s \leq (\log x)^C$, and every $a \in (\mathbb{Z}/s\mathbb{Z})^*$,*

$$|\Delta_{\Lambda_q^b}(y; s, a)| \ll_{A, C} \frac{x}{(\log x)^{A+2C+1}}.$$

Proof. Put $A' := A + 2C + 1$. By Vaughan's identity (Theorem 6.5) and linearity of restriction, $\Lambda_q^b = \Lambda_q - \Lambda_q^\# - (\Lambda_{\leq U})_q$, so by Lemma 3.8 it suffices to bound the three discrepancies separately by $\ll_{A, C} x/(\log x)^{A'}$. Note $y \geq \sqrt{x} \geq 2$ and $\log y \geq \frac{1}{2} \log x \geq 8$ (Lemma 3.14).

Λ_q . Since $(\log y)^2 \geq \frac{1}{4}(\log x)^2 \geq \log x$, we have $s \leq (\log x)^C \leq (\log y)^{2C}$, so Lemma 9.7 applies with $C_2 = 2C$. By Lemma 4.5, $y/(\log y)^{A'} \ll_{A'} x/(\log x)^{A'}$, and by Lemma 5.9 (first bound with $j = 2$, $N = A'$), $(\log x)^2 \ll_{A'} x/(\log x)^{A'}$.

$\Lambda_q^\#$. The reduction of a modulo s is a unit and $q \leq \sqrt{x} \leq x$, so Theorem 8.17 with $r = q$ gives $|\Delta_{\Lambda_q^\#}(y; s, a)| \ll \tau(q) UV \log x \leq \tau(q) \sqrt{x} \log x$, and the third bound of Lemma 5.9 with $N = A'$ gives $\ll_{A'} x/(\log x)^{A'}$.

$(\Lambda_{\leq U})_q$. As in Lemma 7.2 (the restricted function is still non-negative with ℓ^1 norm at most $U \log x$), $|\Delta_{(\Lambda_{\leq U})_q}(y; s, a)| \leq 2U \log x \leq 2\sqrt{x} \log x$, and the second bound of Lemma 5.9 with $j = 1$, $N = A'$ gives $\ll_{A'} x/(\log x)^{A'}$. $\square$

Theorem 9.9 (Small conductors). *Let $A, C \in \mathbb{N}$. For every parameter datum, every real $\sqrt{x} \leq y \leq x$, every integer $1 \leq q \leq \sqrt{x}$ and every $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$\left| \sum_{\substack{d|q \\ 1 < d \leq (\log x)^C}} \sum_{s|d} \mu(d/s) \varphi(s) \Delta_{\Lambda_q^b}(y; s, a) \right| \ll_{A,C} \frac{x}{(\log x)^{A+1}}.$$

Proof. Push the absolute value inside and use $|\mu(d/s)| \leq 1$. For $s \mid d \mid q$ the reduction of a modulo s is a unit and $s \leq d \leq (\log x)^C$, so Theorem 9.8 bounds each discrepancy by $\ll_{A,C} x/(\log x)^{A+2C+1}$. Since $\sum_{s|d} \varphi(s) = d$, the left-hand side is

$$\ll_{A,C} \frac{x}{(\log x)^{A+2C+1}} \sum_{\substack{d|q \\ 1 < d \leq (\log x)^C}} d \leq \frac{x}{(\log x)^{A+2C+1}} \cdot (\log x)^C \cdot (\log x)^C,$$

because there are at most $(\log x)^C$ such d , each at most $(\log x)^C$. $\square$

9.4 Reduction to the quantities T_r

Lemma 9.10 (Maximal form of the conductor decomposition). *Let $A, C \in \mathbb{N}$. For every parameter datum and every integer $1 \leq q \leq \sqrt{x}$,*

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^b}(y; q, a)| \leq \frac{1}{\varphi(q)} \sum_{\substack{d|q \\ d > (\log x)^C}} \sum_{\xi \bmod d}^* \max_{\sqrt{x} \leq y \leq x} S_{q/d}(y, \xi) + O_{A,C} \left(\frac{1}{\varphi(q)} \frac{x}{(\log x)^{A+1}} \right).$$

Proof. By Lemma 3.18 it suffices to bound $|\Delta_{\Lambda^b}(y; q, a)|$ for each unit a and each $\sqrt{x} \leq y \leq x$. Apply Theorem 9.6, bound the first term by Theorem 9.9, and bound each $S_{q/d}(y, \xi)$ in the second term by its maximum over y (Lemma 3.18). $\square$

Theorem 9.11 (Reduction of the Type II estimate to T_r). *Let $A, C \in \mathbb{N}$. For every parameter datum and every real $Q \leq \sqrt{x}$,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^b}(y; q, a)| \leq \sum_{r \leq Q} \frac{T_r^{(C)}(Q)}{\varphi(r)} + O_{A,C} \left(\frac{x}{(\log x)^A} \right).$$

Proof. Sum Lemma 9.10 over $1 \leq q \leq Q$ (each such q satisfies $q \leq \sqrt{x}$).

The error term. Its total is $\ll_{A,C} x(\log x)^{-A-1} \sum_{q \leq Q} 1/\varphi(q)$, and Theorem 5.6 (applicable as $Q \leq \sqrt{x} \leq x$) bounds the sum by $\ll \log x$, giving $\ll_{A,C} x/(\log x)^A$.

The main term. For $d \mid q$ write $q = dr$. Since $\varphi(d)\varphi(r) \leq \varphi(dr)$ (super-multiplicativity of φ), $1/\varphi(q) \leq \varphi(d)^{-1}\varphi(r)^{-1}$. Hence the main term is at most

$$\sum_{q \leq Q} \sum_{\substack{d|q \\ d > (\log x)^C}} \frac{1}{\varphi(r)} \frac{1}{\varphi(d)} \sum_{\xi \bmod d}^* \max_y S_r(y, \xi), \quad r = q/d.$$

The map $(q, d) \mapsto (r, d) = (q/d, d)$ is injective, and its image is contained in the set of pairs with $1 \leq r \leq Q$ and $(\log x)^C < d \leq Q/r$. Since all terms are non-negative, the sum is at most the sum over that larger set, which is $\sum_{r \leq Q} T_r^{(C)}(Q)/\varphi(r)$ by definition of T_r (Definition 9.2). $\square$

9.5 A smoothed Perron formula

The remaining input is a maximal version of the large sieve for Dirichlet convolutions (Theorem 9.37). The textbook proof uses the truncated Perron formula; we use instead a smoothed cutoff $\tilde{1}_\varepsilon$, whose Mellin transform is $s^{-1}\mathcal{M}[\nu](\varepsilon s)$ for a fixed bump function ν . The point is that for $y = K + \frac{1}{2}$ a half-integer and $\varepsilon = 1/(6 \log 2 \cdot x)$, the smoothed cutoff coincides with the sharp cutoff at every integer (Theorem 9.31), so the smoothed bilinear sums $T_\varepsilon(y, \chi)$ are *equal* to the partial sums we want to bound, and no error term needs to be estimated. In this section $x \geq 1$ and $Q \geq 1$ are arbitrary reals, (f, g) is a supported pair and ν a bump function; no parameter datum is involved.

Inputs from the PrimeNumberTheoremAnd library

The smoothed cutoff and its basic properties are taken from the PrimeNumberTheoremAnd library (module MellinCalculus). We record the statements we use; the proofs are in that library and are not reproduced here. Throughout, $\nu : \mathbb{R} \rightarrow \mathbb{R}$ is a C^1 function supported in $[1/2, 2]$.

Theorem 9.12 (Existence of a bump function). *There exists a C^∞ function $\nu : \mathbb{R} \rightarrow \mathbb{R}$, supported in $[1/2, 2]$, with $\nu \geq 0$ everywhere and $\int_0^\infty \nu(t) \frac{dt}{t} = 1$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Definition 9.13 (The smoothed cutoff $\tilde{1}_\varepsilon$). For $\varepsilon > 0$, the smoothed cutoff $\tilde{1}_\varepsilon : \mathbb{R} \rightarrow \mathbb{R}$ is the multiplicative (Mellin) convolution of the indicator of $(0, 1]$ with the spike $t \mapsto \nu(t^{1/\varepsilon})/\varepsilon$:

$$\tilde{1}_\varepsilon(u) := \int_0^\infty 1_{(0,1]}(t) \frac{1}{\varepsilon} \nu((u/t)^{1/\varepsilon}) \frac{dt}{t}.$$

Lemma 9.14 (The smoothed cutoff equals 1 below $1 - (\log 2)\varepsilon$). *Assume $\int_0^\infty \nu(t) dt/t = 1$. For $\varepsilon > 0$ and $0 < u \leq 1 - (\log 2)\varepsilon$ we have $\tilde{1}_\varepsilon(u) = 1$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Lemma 9.15 (The smoothed cutoff vanishes above $1 + 2(\log 2)\varepsilon$). *For $0 < \varepsilon < 1$ and $u \geq 1 + 2(\log 2)\varepsilon$ we have $\tilde{1}_\varepsilon(u) = 0$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Lemma 9.16 (Non-negativity of the smoothed cutoff). *Assume $\nu \geq 0$ on $(0, \infty)$. Then $\tilde{1}_\varepsilon(u) \geq 0$ for all $\varepsilon > 0$ and $u > 0$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Lemma 9.17 (The smoothed cutoff is at most 1). *Assume $\nu \geq 0$ on $(0, \infty)$ and $\int_0^\infty \nu(t) dt/t = 1$. Then $\tilde{1}_\varepsilon(u) \leq 1$ for all $\varepsilon > 0$ and $u > 0$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Lemma 9.18 (Continuity of the smoothed cutoff). *Assume $\nu \geq 0$ on $(0, \infty)$. For $\varepsilon > 0$, $\tilde{1}_\varepsilon$ is continuous at every $u > 0$.*

Proof. Proved in PrimeNumberTheoremAnd. □

Lemma 9.19 (Mellin transform of the smoothed cutoff). *For $\varepsilon > 0$ and $\Re s > 0$,*

$$\mathcal{M}[\tilde{1}_\varepsilon](s) = \frac{1}{s} \mathcal{M}[\nu](\varepsilon s), \quad \text{where } \mathcal{M}[f](s) := \int_0^\infty u^{s-1} f(u) du.$$

Proof. Proved in PrimeNumberTheoremAnd. $\square$

Lemma 9.20 (Decay of the Mellin transform of the smoothed cutoff). *There is a constant $C > 0$, depending only on ν , such that for all $\sigma_1 > 0$, all s with $\sigma_1 \leq \Re s \leq 2$ and all $0 < \varepsilon < 1$,*

$$\|\mathcal{M}[\tilde{1}_\varepsilon](s)\| \leq \frac{C}{\varepsilon \|s\|^2}.$$

Proof. Proved in PrimeNumberTheoremAnd. $\square$

Lemma 9.21 (Integration by parts on $(0, \infty)$). *Let $f, g : \mathbb{R} \rightarrow \mathbb{C}$ be differentiable on $(0, \infty)$ with fg' and $f'g$ integrable on $(0, \infty)$ and $f(u)g(u) \rightarrow 0$ as $u \rightarrow 0^+$ and as $u \rightarrow \infty$. Then $\int_0^\infty fg' = -\int_0^\infty f'g$.*

Proof. Proved in PrimeNumberTheoremAnd. $\square$

The smoothed bilinear sums

Definition 9.22 (Bump functions and the smoothed cutoff). A *bump function* is a function $\nu : \mathbb{R} \rightarrow \mathbb{R}$ that is C^∞ , supported in $[1/2, 2]$, non-negative, and normalised by $\int_0^\infty \nu(t) \frac{dt}{t} = 1$. Such functions exist (Theorem 9.12); one is fixed once and for all, and the implied constants in this section may depend on this choice. For $\varepsilon > 0$ we write $\tilde{1}_\varepsilon$ for the smoothed cutoff of Definition 9.13 built from ν ; it satisfies $0 \leq \tilde{1}_\varepsilon \leq 1$ (Lemma 9.16), equals 1 on $(0, 1 - (\log 2)\varepsilon]$ and 0 on $[1 + 2(\log 2)\varepsilon, \infty)$ when $\varepsilon < 1$ (Lemma 9.14), and has Mellin transform $\mathcal{M}[\tilde{1}_\varepsilon](s) = s^{-1} \mathcal{M}[\nu](\varepsilon s)$ for $\Re s > 0$ (Lemma 9.19).

Definition 9.23 (Supported pairs of arithmetic functions). A *supported pair* consists of complex arithmetic functions f, g and real numbers $M, N > 0$ with $f(n) = 0$ for $n > M$ and $g(n) = 0$ for $n > N$. For such a pair, a character χ and $s \in \mathbb{C}$ we write

$$F_s(\chi) := \sum_{m \leq M} f(m) \chi(m) m^{-s}, \quad G_s(\chi) := \sum_{n \leq N} g(n) \chi(n) n^{-s}$$

for the associated Dirichlet polynomials, and $\|f\|_2 := (\sum_{m \leq M} |f(m)|^2)^{1/2}$, $\|g\|_2 := (\sum_{n \leq N} |g(n)|^2)^{1/2}$.

Definition 9.24 (The smoothed bilinear sums $T_\varepsilon(y, \chi)$). For a supported pair (f, g) , a bump function ν , $\varepsilon > 0$, $y > 0$ and a Dirichlet character χ , set

$$T_\varepsilon(y, \chi) := \sum_{m \leq M} \sum_{n \leq N} f(m) \chi(m) g(n) \chi(n) \tilde{1}_\varepsilon\left(\frac{mn}{y}\right).$$

Lemma 9.25 (The Mellin transform of a bump is bounded on strips). *Let $\nu \in C^1(\mathbb{R})$ be supported in $[1/2, 2]$ and let $\sigma_1 \leq \sigma_2$ with $\sigma_2 \geq 0$. Then there is a constant C such that $\|\mathcal{M}[\nu](s)\| \leq C \|s\|^{-1}$ for all $s \neq 0$ with $\sigma_1 \leq \Re s \leq \sigma_2$; consequently $\mathcal{M}[\nu]$ is bounded on the whole strip $\{\sigma_1 \leq \Re s \leq \sigma_2\}$.*

Proof. Integration by parts (valid since ν has compact support in $(0, \infty)$) gives $\mathcal{M}[\nu](s) = \int_0^\infty \nu(u) u^{s-1} du = -\frac{1}{s} \int_0^\infty \nu'(u) u^s du$ for $s \neq 0$. On $[1/2, 2]$ we have $|u^s| = u^{\Re s} \leq \max(2^{\sigma_2}, 2^{-\sigma_1})$, and $|\nu'|$ is bounded by its maximum on $[1/2, 2]$; the interval has length $3/2$. This gives the first bound with $C = \frac{3}{2} \max |\nu'| \cdot \max(2^{\sigma_2}, 2^{-\sigma_1})$. For boundedness on the strip, note that $\mathcal{M}[\nu]$ is entire (the Mellin integral of a compactly supported C^1 function converges for every s), hence bounded near $s = 0$; away from a small ball around 0 the first bound applies. $\square$

Lemma 9.26 (Mellin inversion for the smoothed cutoff). *Let ν be a bump function (Definition 9.22; only $\nu \in C^1$ is needed), $0 < \varepsilon < 1$ and $0 < \sigma \leq 2$. Then $t \mapsto \mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)$ is integrable on $\mathbb{R}$ and, for every $u > 0$,*

$$\tilde{1}_\varepsilon(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u^{-(\sigma+it)} \mathcal{M}[\tilde{1}_\varepsilon](\sigma + it) dt.$$

Proof. Integrability: by the bound $\|\mathcal{M}[\tilde{1}_\varepsilon](s)\| \ll_\nu 1/(\varepsilon\|s\|^2)$ on the strip $\sigma/2 \leq \Re s \leq 2$ (Lemma 9.20), the integrand is dominated by a constant multiple of $(\sigma^2 + t^2)^{-1}$, which is integrable; measurability follows from continuity of $\mathcal{M}[\tilde{1}_\varepsilon]$ on $\Re s > 0$. The inversion formula is then Mathlib's Mellin inversion theorem (`mellinInv_mellin_eq`), whose hypotheses are: convergence of the Mellin integral at σ (`Smooth1MellinConvergent`), vertical integrability of the transform (just shown), and continuity of $\tilde{1}_\varepsilon$ at u (`Smooth1ContinuousAt`). $\square$

Lemma 9.27 (Two bounds for the Mellin transform of $\tilde{1}_\varepsilon$). *Let ν be a bump function. For all $0 < \varepsilon < 1$, all $0 < \sigma \leq 2$ and all $t \in \mathbb{R}$, with $s = \sigma + it$,*

$$\|\mathcal{M}[\tilde{1}_\varepsilon](s)\| \ll \frac{1}{\|s\|} \quad \text{and} \quad \|\mathcal{M}[\tilde{1}_\varepsilon](s)\| \ll \frac{1}{\varepsilon\|s\|^2},$$

the implied constants depending only on ν .

Proof. For the first bound use $\mathcal{M}[\tilde{1}_\varepsilon](s) = s^{-1} \mathcal{M}[\nu](\varepsilon s)$ (Definition 9.22) and the boundedness of $\mathcal{M}[\nu]$ on the strip $0 \leq \Re \leq 2$ (Lemma 9.25); note $\Re(\varepsilon s) = \varepsilon\sigma \in (0, 2)$. The second bound is Lemma 9.20, applied with lower edge $\sigma_1 = \sigma$; its constant depends only on ν . (Only these two bounds are needed; in particular no rapid decay of $\mathcal{M}[\nu]$ is used.) $\square$

Lemma 9.28 (The kernel integral J). *Let ν be a bump function. Let $x \geq 1$, $\sigma := 1/\log(x+1)$ and $\varepsilon := \frac{1}{6\log 2} \cdot \frac{1}{x}$ (so that $0 < \varepsilon < 1$ and $0 < \sigma \leq 2$). Then*

$$J := \int_{-\infty}^{\infty} \|\mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)\| dt \ll \log(x+1),$$

the implied constant depending only on ν .

Proof. Write $L = \log(x+1) \geq \log 2$ and split the integral at $|t| = T := 1/\varepsilon$.

Centre $|t| \leq T$. By the first bound of Lemma 9.27 and the elementary inequality $\|\sigma + it\|^{-1} \leq \sqrt{2}(\sigma + |t|)^{-1}$ (equivalent to $(\sigma - |t|)^2 \geq 0$), the integrand is $\ll (\sigma + |t|)^{-1}$, and $\int_{-T}^T (\sigma + |t|)^{-1} dt = 2 \log((\sigma + T)/\sigma) = 2 \log(1 + 6 \log 2 \cdot xL)$. Since $1 + 6 \log 2 \cdot xL \leq (1+x)(1 + 6 \log 2 \cdot L)$ and $\log(1+u) \leq u$, this is $\ll L$.

Tails $|t| > T$. By the second bound of Lemma 9.27 and $t^2 \leq \|\sigma + it\|^2$, the integrand is $\ll \varepsilon^{-1} t^{-2}$, and $\int_{|t|>T} t^{-2} dt = 2/T = 2\varepsilon$. The tails contribute $\ll 1 \ll L$. $\square$

Lemma 9.29 (Reduction of $\max_{y \leq x}$ to half-integers). *For $x \geq 1$ and $f : \mathbb{N} \rightarrow \mathbb{C}$,*

$$\sup_{1 \leq y \leq x} \left\| \sum_{n \leq y} f(n) \right\| = \max_{1 \leq K \leq \lfloor x \rfloor} \left\| \sum_{n \leq K+1/2} f(n) \right\|.$$

Consequently, for every supported pair (f, g) (Definition 9.23) and $Q \geq 1$,

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x} \left\| \sum_{n \leq y} (f * g)(n) \chi(n) \right\| = \sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \max_{1 \leq K \leq \lfloor x \rfloor} \left\| \sum_{n \leq K+1/2} (f * g)(n) \chi(n) \right\|.$$

Proof. The partial sum $\sum_{n \leq y} f(n)$ depends only on $\lfloor y \rfloor$, and $\lfloor K + 1/2 \rfloor = K$. As y ranges over $[1, x]$, $\lfloor y \rfloor$ ranges over $\{1, \dots, \lfloor x \rfloor\}$, so both sides are the maximum of the same finite set of numbers. The second statement follows by applying the first inside each summand. $\square$

Lemma 9.30 (Partial sums of a convolution as a double sum). *For a supported pair (f, g) (Definition 9.23), a character χ and every real y ,*

$$\sum_{n \leq y} (f * g)(n) \chi(n) = \sum_{m \leq M} \sum_{n \leq N} 1_{mn \leq y} f(m) \chi(m) g(n) \chi(n).$$

Proof. The hyperbola identity $\sum_{n \leq y} (F * G)(n) = \sum_{m \leq y} F(m) \sum_{n \leq y/m} G(n)$ with $F = f\chi$, $G = g\chi$ (using complete multiplicativity of χ) writes the left side as $\sum_{m \leq y} \sum_{n \leq y/m} f(m) \chi(m) g(n) \chi(n)$. Enlarging both ranges to $m \leq \lfloor y \rfloor + \lfloor M \rfloor + \lfloor N \rfloor$ and inserting the indicator $1_{mn \leq y}$ changes nothing, and then the support conditions on f and g allow one to shrink the ranges to $m \leq M$, $n \leq N$. $\square$

Theorem 9.31 (The smoothed sum is sharp at half-integers). *Let $x \geq 1$, let (f, g) be a supported pair, ν a bump function, χ a Dirichlet character, and let $0 < \varepsilon \leq \frac{1}{6 \log 2} \cdot \frac{1}{x}$. Then for every integer $0 \leq K \leq x$,*

$$T_\varepsilon(K + \tfrac{1}{2}, \chi) = \sum_{m \leq M} \sum_{n \leq N} 1_{mn \leq K+1/2} f(m) \chi(m) g(n) \chi(n).$$

In words: with this choice of ε the smoothed cutoff $\tilde{1}_\varepsilon(mn/(K + \frac{1}{2}))$ agrees with the sharp cutoff $1_{mn \leq K+1/2}$ at every integer mn , so no error term arises.

Proof. Compare the two sums termwise; put $u = mn/(K + \frac{1}{2})$ with $mn \geq 1$ an integer.

If $mn \leq K + \frac{1}{2}$ then $mn \leq K$, so $u \leq K/(K + \frac{1}{2}) = 1 - \frac{1}{2K+1}$. Since $(\log 2)\varepsilon \leq \frac{1}{6x} \leq \frac{1}{2K+1}$ (as $2K + 1 \leq 6x$ for $K \leq x$, $x \geq 1$), we get $u \leq 1 - (\log 2)\varepsilon$ and $\tilde{1}_\varepsilon(u) = 1$ (Definition 9.22).

If $mn > K + \frac{1}{2}$ then $mn \geq K + 1$, so $u \geq (K + 1)/(K + \frac{1}{2}) = 1 + \frac{1}{2K+1}$. Since $2(\log 2)\varepsilon \leq \frac{1}{3x} \leq \frac{1}{2K+1}$ (as $4K + 2 \leq 6x$), we get $u \geq 1 + 2(\log 2)\varepsilon$, and also $\varepsilon < 1$, so $\tilde{1}_\varepsilon(u) = 0$ (Definition 9.22). $\square$

Theorem 9.32 (Mellin representation of T_ε). *Let (f, g) be a supported pair, ν a bump function, $0 < \varepsilon < 1$, $0 < \sigma \leq 2$, $y \geq 1$ and χ a Dirichlet character. Then*

$$T_\varepsilon(y, \chi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_{\sigma+it}(\chi) G_{\sigma+it}(\chi) y^{\sigma+it} \mathcal{M}[\tilde{1}_\varepsilon](\sigma + it) dt,$$

and consequently

$$\|T_\varepsilon(y, \chi)\| \leq \frac{y^\sigma}{2\pi} \int_{-\infty}^{\infty} \|F_{\sigma+it}(\chi)\| \|G_{\sigma+it}(\chi)\| \|\mathcal{M}[\tilde{1}_\varepsilon](\sigma + it)\| dt,$$

where the last integrand is integrable.

Proof. Apply Lemma 9.26 to each term of $T_\varepsilon(y, \chi)$ at $u = mn/y > 0$:

$$T_\varepsilon(y, \chi) = \sum_{m \leq M} \sum_{n \leq N} f(m)\chi(m)g(n)\chi(n) \cdot \frac{1}{2\pi} \int_{\mathbb{R}} \left(\frac{mn}{y}\right)^{-(\sigma+it)} \mathcal{M}[\tilde{1}_\varepsilon](\sigma+it) dt.$$

Each integrand is a continuous function of constant modulus $\|f(m)\chi(m)\|m^{-\sigma}\|g(n)\chi(n)\|n^{-\sigma}y^\sigma$ times the integrable function $\mathcal{M}[\tilde{1}_\varepsilon](\sigma+it)$, hence integrable, so the finite double sum can be moved inside the integral. Using $(mn/y)^{-s} = m^{-s}n^{-s}y^s$ for positive reals and collecting the sums gives $F_s(\chi)G_s(\chi)y^s$. The inequality follows by taking norms inside the integral and using $|y^{\sigma+it}| = y^\sigma$; integrability of the majorant follows since $\|F_{\sigma+it}\|\|G_{\sigma+it}\|$ is continuous and bounded by $\sum_{m \leq M} |f(m)|m^{-\sigma} \cdot \sum_{n \leq N} |g(n)|n^{-\sigma}$. $\square$

9.6 The large sieve for convolutions

Lemma 9.33 (The large sieve for a Dirichlet polynomial). *Let $Q \geq 1$, $\sigma > 0$, $t \in \mathbb{R}$, $h : \mathbb{N} \rightarrow \mathbb{C}$ and $P \in \mathbb{R}$. Then*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \left\| \sum_{m \leq P} h(m)\chi(m)m^{-(\sigma+it)} \right\|^2 \ll (P+Q^2) \sum_{m \leq P} |h(m)|^2,$$

with the implied constant of Assumption 4.6.

Proof. If $P < 1$ the left side is 0. Otherwise apply Assumption 4.6 with $H = 0$, $N = \lfloor P \rfloor$ and $c_n = h(n)n^{-(\sigma+it)}$ for $1 \leq n \leq N$. Since $|c_n|^2 = |h(n)|^2 n^{-2\sigma} \leq |h(n)|^2$ for $n \geq 1$, and $\lfloor P \rfloor \leq P$, the right-hand side of the large sieve inequality is $\ll (P+Q^2) \sum_{m \leq P} |h(m)|^2$. $\square$

Theorem 9.34 (Cauchy–Schwarz and the large sieve twice). *Let (f, g) be a supported pair (Definition 9.23), $Q \geq 1$, $\sigma > 0$ and $t \in \mathbb{R}$. Then*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \|F_{\sigma+it}(\chi)\| \|G_{\sigma+it}(\chi)\| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2,$$

with the implied constant of Assumption 4.6; in particular it is uniform in t and σ .

Proof. By the Cauchy–Schwarz inequality with weights $w_{q,\chi} = q/\varphi(q) \geq 0$ over the pairs (q, χ) ,

$$\sum_{q,\chi} w_{q,\chi} \|F\| \|G\| \leq \left(\sum_{q,\chi} w_{q,\chi} \|F\|^2 \right)^{1/2} \left(\sum_{q,\chi} w_{q,\chi} \|G\|^2 \right)^{1/2}.$$

By Lemma 9.33 (with $h = f$, $P = M$, and with $h = g$, $P = N$) the two factors are $\ll \sqrt{M+Q^2} \|f\|_2$ and $\ll \sqrt{N+Q^2} \|g\|_2$. Finally $\sqrt{M+Q^2} \sqrt{N+Q^2} \leq (\sqrt{M}+Q)(\sqrt{N}+Q) = \sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2$, using $\sqrt{a+Q^2} \leq \sqrt{a}+Q$ for $a, Q \geq 0$. $\square$

Theorem 9.35 (The smoothed bilinear sums on average). *Let (f, g) be a supported pair, ν a bump function, $Q \geq 1$, $x \geq 1$ and $\varepsilon := \frac{1}{6 \log 2} \cdot \frac{1}{x}$. Then*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x+1} \|T_\varepsilon(y, \chi)\| \ll (\sqrt{NM} + \sqrt{M}Q + \sqrt{N}Q + Q^2) \|f\|_2 \|g\|_2 \log(x+1),$$

and the same bound holds with the supremum replaced by $\max_{1 \leq K \leq \lfloor x \rfloor} \|T_\varepsilon(K + \frac{1}{2}, \chi)\|$. The implied constant depends only on ν and on the constant in Assumption 4.6.

Proof. Put $L := \log(x+1) \geq \log 2$ and $\sigma := 1/L$, so that $0 < \sigma \leq 2$ and $(x+1)^\sigma = e$; hence $y^\sigma \leq e$ for $1 \leq y \leq x+1$. Also $\varepsilon \leq 1/(6 \log 2) < 1$. For each pair (q, χ) , Theorem 9.32 gives the y -free majorant

$$\sup_{1 \leq y \leq x+1} \|T_\varepsilon(y, \chi)\| \leq \frac{e}{2\pi} \int_{\mathbb{R}} \|F_{\sigma+it}(\chi)\| \|G_{\sigma+it}(\chi)\| \|\mathcal{M}[\tilde{1}_\varepsilon](\sigma+it)\| dt.$$

Multiply by $q/\varphi(q)$, sum over the finitely many (q, χ) , and interchange the sum with the integral (all integrands are integrable). The resulting integrand is $D(t) \|\mathcal{M}[\tilde{1}_\varepsilon](\sigma+it)\|$ with $D(t) = \sum_{q, \chi} \frac{q}{\varphi(q)} \|F_{\sigma+it}(\chi)\| \|G_{\sigma+it}(\chi)\|$, which by Theorem 9.34 is $\ll (\sqrt{NM} + \sqrt{MQ} + \sqrt{NQ} + Q^2) \|f\|_2 \|g\|_2$ uniformly in t . Pulling this bound out, the remaining integral is $J \ll L$ by Lemma 9.28.

For the discrete version note that $K + \frac{1}{2} \in [1, x+1]$ for $1 \leq K \leq \lfloor x \rfloor$, so the maximum over K is at most the supremum over y . $\square$

Theorem 9.36 (Maximal large sieve for convolutions (smoothed form)). *Let (f, g) be a supported pair (Definition 9.23), ν a bump function, $x \geq 1$ and $Q \geq 1$. Then*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x} \left\| \sum_{n \leq y} (f * g)(n) \chi(n) \right\| \ll (\sqrt{NM} + \sqrt{MQ} + \sqrt{NQ} + Q^2) \|f\|_2 \|g\|_2 \log(x+1).$$

This is Theorem 26.6 of Koukoulopoulos, proved with a smoothed cutoff instead of the truncated Perron formula; the implied constant depends only on the bump ν and on the constant in Assumption 4.6.

Proof. By Lemma 9.29 the supremum over $y \in [1, x]$ may be replaced by the maximum over integers $1 \leq K \leq \lfloor x \rfloor$ of the partial sum at $y = K + \frac{1}{2}$. By Lemma 9.30 that partial sum is $\sum_{m \leq M} \sum_{n \leq N} 1_{mn \leq K+1/2} f(m) \chi(m) g(n) \chi(n)$, which by Theorem 9.31 with $\varepsilon = \frac{1}{6 \log 2} \cdot \frac{1}{x}$ equals $T_\varepsilon(K + \frac{1}{2}, \chi)$ (note $K \leq \lfloor x \rfloor \leq x$). Now apply the discrete form of Theorem 9.35. $\square$

Theorem 9.37 (Maximal large sieve for convolutions). *Let $M, N \in \mathbb{N}$ and let f, g be real arithmetic functions with $f(n) = 0$ for $n > M$ and $g(n) = 0$ for $n > N$. Then for all real $x \geq 2$ and $Q \geq 1$,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \sup_{1 \leq y \leq x} \left| \sum_{n \leq y} (f * g)(n) \chi(n) \right| \ll (\sqrt{NM} + \sqrt{MQ} + \sqrt{NQ} + Q^2) (\log x) \left(\sum_{n \leq M} f(n)^2 \right)^{1/2} \left(\sum_{n \leq N} g(n)^2 \right)^{1/2},$$

with an implied constant depending only on the fixed bump function and on the constant in Assumption 4.6.

Proof. If $M = 0$ or $N = 0$ then $f * g = 0$ and the left side vanishes. Otherwise (f, g) , viewed as complex arithmetic functions, form a supported pair with the real parameters $M, N > 0$, and Theorem 9.36 gives the bound with $\log(x+1)$ in place of $\log x$. Finally $\log(x+1) \leq \log(x^2) = 2 \log x$ for $x \geq 2$. $\square$

9.7 Dyadic decomposition and the bound for T_r

Lemma 9.38 (Dyadic decomposition of Λ^b). *For every parameter datum and every integer $n \leq x$,*

$$\Lambda^b(n) = \sum_{\substack{j \in \mathbb{N} \\ U < 2^j \leq 2x/V}} (f_j * g_j)(n), \quad f_j := (\Lambda_{>U} * 1) 1_{(2^{j-1}, 2^j]}, \quad g_j := \mu \cdot 1_{(V, x/2^{j-1}]}. \quad \square$$

Proof. Expand $\Lambda^b(n) = \sum_{de=n} (\Lambda_{>U} * 1)(d) \mu_{>V}(e)$ (Definition 6.4) and, on the right, $\sum_j (f_j * g_j)(n) = \sum_{de=n} \sum_j f_j(d) g_j(e)$. Fix a pair $de = n$. If $d \leq U$ then $(\Lambda_{>U} * 1)(d) = \sum_{k|d} \Lambda_{>U}(k) = 0$ and every $f_j(d) = 0$; if $e \leq V$ then $\mu_{>V}(e) = 0$ and every $g_j(e) = 0$. Otherwise $d > U \geq 1$ and $e > V$, and there is a unique $j_0 \geq 1$ with $2^{j_0-1} < d \leq 2^{j_0}$. It lies in the summation range, since $U < d \leq 2^{j_0}$ and $2^{j_0} < 2d \leq 2n/e \leq 2x/V$; moreover $e = n/d \leq x/d < x/2^{j_0-1}$, so $f_{j_0}(d) g_{j_0}(e) = (\Lambda_{>U} * 1)(d) \mu(e) = (\Lambda_{>U} * 1)(d) \mu_{>V}(e)$, while $f_j(d) = 0$ for $j \neq j_0$. $\square$

Theorem 9.39 (The large sieve bound for S_r). *For every parameter datum, every $r \in \mathbb{N}$ and every real $Q \geq 2$,*

$$\sum_{q \leq Q} \sum_{\chi \bmod q}^* \frac{q}{\varphi(q)} \max_{\sqrt{x} \leq y \leq x} S_r(y, \chi) \ll \left(x + \frac{Qx}{\sqrt{U}} + \frac{Qx}{\sqrt{V}} + Q^2 \sqrt{x} \right) (\log x)^3.$$

Proof. Dyadic decomposition. Restricting Lemma 9.38 to integers coprime to r and using that restriction is a convolution homomorphism (Lemma 8.6), $\Lambda_r^b(n) = \sum_j ((f_j)_r * (g_j)_r)(n)$ for $n \leq x$, the sum being over the set $\mathcal{J} = \{j : U < 2^j \leq 2x/V\}$. Hence, for $\sqrt{x} \leq y \leq x$ (so $1 \leq y \leq x$),

$$S_r(y, \chi) \leq \sum_{j \in \mathcal{J}} \sup_{1 \leq z \leq x} \left| \sum_{n \leq z} ((f_j)_r * (g_j)_r)(n) \chi(n) \right| =: \sum_{j \in \mathcal{J}} \Gamma_{r,j}(\chi),$$

and the same bound holds for the maximum over y .

One dyadic block. Fix $j \in \mathcal{J}$ and put $M_j := 2^j$, $N_j := \lfloor x/2^{j-1} \rfloor = \lfloor 2x/M_j \rfloor$. The function $(f_j)_r$ is supported on $[1, M_j]$ and, since $0 \leq (\Lambda_{>U} * 1)(n) \leq \sum_{k|n} \Lambda(k) = \log n \leq 2 \log x$ for $n \leq 2^j \leq x^2$, it satisfies $\|(f_j)_r\|_2^2 \leq M_j (2 \log x)^2$. The function $(g_j)_r$ is supported on $[1, N_j]$ with $|(g_j)_r| \leq 1$, so $\|(g_j)_r\|_2^2 \leq N_j \leq 2x/M_j$. Thus $\|(f_j)_r\|_2 \|(g_j)_r\|_2 \ll \log x \sqrt{x}$. By Theorem 9.37 (with $x \geq 2$, $Q \geq 1$),

$$\sum_{q \leq Q} \sum_{\chi}^* \frac{q}{\varphi(q)} \Gamma_{r,j}(\chi) \ll (\sqrt{N_j M_j} + \sqrt{M_j} Q + \sqrt{N_j} Q + Q^2) (\log x)^2 \sqrt{x}.$$

Now $\sqrt{N_j M_j} \leq \sqrt{2x}$; $\sqrt{M_j} \sqrt{x} \ll x/\sqrt{V}$ because $M_j \leq 2x/V$; and $\sqrt{N_j} \sqrt{x} \ll x/\sqrt{M_j} \leq x/\sqrt{U}$ because $M_j > U$. Hence each block contributes $\ll (\log x)^2 (x + Qx/\sqrt{U} + Qx/\sqrt{V} + Q^2 \sqrt{x})$, uniformly in j .

Number of blocks. $\#\mathcal{J} \leq \log_2(2x) \ll \log x$, since $\log x \geq \log 2$.

Multiplying gives the claim. Note that the savings $1/\sqrt{U}$ and $1/\sqrt{V}$ are obtained termwise from $U < 2^j \leq 2x/V$ rather than by summing a geometric series; this costs one factor $\log x$ (the number of blocks). $\square$

Theorem 9.40 (The bound for T_r). *For every parameter datum, every integer $C \geq 3$, every $r \in \mathbb{N}$ and every real $2 \leq Q \leq x$,*

$$T_r^{(C)}(Q) \ll \frac{x}{(\log x)^{C-3}} + \frac{x(\log x)^4}{\sqrt{U}} + \frac{x(\log x)^4}{\sqrt{V}} + \frac{Q\sqrt{x}(\log x)^3}{r},$$

with an implied constant independent of C and r .

Proof. Write $B(d) := \sum_{\xi \bmod d}^* \frac{d}{\varphi(d)} \max_y S_r(y, \xi) \geq 0$, so that $T_r(Q) = \sum_{(\log x)^C < d \leq Q/r} B(d)/d$. If the range is empty (in particular if $r = 0$) there is nothing to prove; otherwise $r \geq 1$ and $Q/r \leq x$.

Dyadic blocks in d . Group the d in the range by $j = \lfloor \log_2 d \rfloor$, which ranges over $[j_L, j_U]$ with $j_L = \lfloor \log_2 \lfloor (\log x)^C \rfloor \rfloor$ and $j_U = \lfloor \log_2 \lfloor Q/r \rfloor \rfloor$; each d in the block j satisfies $2^j \leq d < 2^{j+1}$, so $1/d \leq 2^{-j}$. Hence

$$T_r(Q) \leq \sum_{j=j_L}^{j_U} 2^{-j} \sum_{d \leq 2^{j+1}} B(d) \ll \sum_{j=j_L}^{j_U} 2^{-j} \left(x + \frac{2^{j+1}x}{\sqrt{U}} + \frac{2^{j+1}x}{\sqrt{V}} + 4^{j+1}\sqrt{x} \right) (\log x)^3$$

by Theorem 9.39 applied with $Q' = 2^{j+1} \geq 2$.

Summing over j . Since $(\log x)^C < 2^{j_L+1}$, we have $\sum_{j \geq j_L} 2^{-j} \leq 2 \cdot 2^{-j_L} \leq 4/(\log x)^C$, so the first term contributes $\ll x(\log x)^{3-C}$. The two middle terms are independent of j and the number of j is at most $j_U + 1 \leq \log_2(2Q/r) \ll \log x$ (as $2^{j_U} \leq Q/r \leq x$), contributing $\ll x(\log x)^4/\sqrt{U}$ and the same with V . Finally $\sum_{j \leq j_U} 2^{j+1} \leq 4 \cdot 2^{j_U} \leq 4Q/r$, so the last term contributes $\ll Q\sqrt{x}(\log x)^3/r$. $\square$

9.8 The Type II estimate

Theorem 9.41 (The Type II part on average). *For every parameter datum, every $A \in \mathbb{N}$ and every real $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_{\Lambda^*}(y; q, a)| \ll_A \frac{x}{(\log x)^A}.$$

Proof. Since $\log x \geq 1$, $Q \leq \sqrt{x} \leq x$. Apply Theorem 9.11 with $C = A + 4$; the error term is $\ll_A x/(\log x)^A$, and it remains to bound $\sum_{r \leq Q} T_r^{(A+4)}(Q)/\varphi(r)$.

Small Q . If $Q < 2$ then for every $r \geq 1$ we have $Q/r < 2 \leq 16 \leq (\log x)^{A+4}$ (Lemma 3.14), so the range of d in T_r is empty and the main term vanishes.

$Q \geq 2$. By Theorem 9.40 with $C = A + 4$ (so $C - 3 = A + 1$ and $Q \leq x$),

$$\sum_{r \leq Q} \frac{T_r(Q)}{\varphi(r)} \ll \left(\frac{x}{(\log x)^{A+1}} + \frac{x(\log x)^4}{\sqrt{U}} + \frac{x(\log x)^4}{\sqrt{V}} \right) \sum_{r \leq Q} \frac{1}{\varphi(r)} + Q\sqrt{x}(\log x)^3 \sum_{r \leq Q} \frac{1}{r\varphi(r)}.$$

By Theorem 5.6, $\sum_{r \leq Q} 1/\varphi(r) \ll \log x$. The first product is then $\ll x/(\log x)^A$. For the second and third, Lemma 5.10 with $n = A + 5$ and $W = U$ (resp. V) gives $(\log x)^{A+5} \ll_A \sqrt{U}$, i.e. $x(\log x)^5/\sqrt{U} \ll_A x/(\log x)^A$. For the last term, Lemma 5.7 bounds the sum by $\ll 1$, and $Q\sqrt{x}(\log x)^3 \leq \sqrt{x}(\log x)^{-A-3}\sqrt{x}(\log x)^3 = x/(\log x)^A$. $\square$

Chapter 10

Assembly

Lemma 10.1 (Splitting Δ_Λ into three parts). *For every parameter datum and every $q \in \mathbb{N}$,*

$$\max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_\Lambda(y; q, a)| \leq \max_{y, a} |\Delta_{\Lambda^\sharp}(y; q, a)| + \max_{y, a} |\Delta_{\Lambda^\flat}(y; q, a)| + \max_{y, a} |\Delta_{\Lambda_{\leq U}}(y; q, a)|.$$

Proof. By Vaughan's identity (Theorem 6.5), $\Lambda = (\Lambda^\sharp + \Lambda^\flat) + \Lambda_{\leq U}$ as functions on $\mathbb{N}$; apply the maximal triangle inequality Lemma 3.20 twice. $\square$

Theorem 10.2 (Bombieri–Vinogradov for Δ_Λ). *For every parameter datum, every $A \in \mathbb{N}$ and every real $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \max_{\substack{\sqrt{x} \leq y \leq x \\ a \in (\mathbb{Z}/q\mathbb{Z})^*}} |\Delta_\Lambda(y; q, a)| \ll_A \frac{x}{(\log x)^A}.$$

Proof. Sum Lemma 10.1 over $q \leq Q$ and bound the three resulting sums by Theorem 8.18, Theorem 9.41 and Theorem 7.3 (the last gives $\ll x/(\log x)^{A+2} \leq x/(\log x)^A$, as $\log x \geq 1$). $\square$

Definition 10.3 (The maximal Chebyshev error). For a real x and $q \in \mathbb{N}$ set

$$E(x; q) := \sup_{1 \leq y \leq x} \sup_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \in [0, \infty].$$

Note that here y ranges over all of $[1, x]$, not only over $[\sqrt{x}, x]$, and that $x \geq 2$ is arbitrary (no parameter datum is involved). For $q \geq 1$, $E(x; q) \ll x$ by Lemma 5.1, so $E(x; q)$ is finite.

Lemma 10.4 (The compact range $x \leq e^{32}$). *Let $A \in \mathbb{N}$ and let $2 \leq x \leq e^{32}$ and $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$ be real. Then*

$$\sum_{q \leq Q} E(x; q) \ll_A \frac{x}{(\log x)^A}.$$

Proof. By Lemma 5.1, $E(x; q) \ll x$ for every $q \geq 1$, so the sum is $\ll Qx$. Since $\sqrt{x} \leq e^{16}$ and $\log x \geq \log 2$, the hypothesis gives $Q \leq e^{16}/(\log 2)^{A+3} \ll_A 1$. Finally $(\log x)^A \leq 32^A$, i.e. $1 \ll_A (\log x)^{-A}$. $\square$

Lemma 10.5 (From Δ_Λ to the Chebyshev error). *Let $A \in \mathbb{N}$. For every parameter datum and every real $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} E(x; q) \ll_A \frac{x}{(\log x)^A}.$$

Proof. Write $L := \log x \geq 1$ and note $Q \leq \sqrt{x} \leq x$ and $\sqrt{x} \geq 2$ (Lemma 3.14).

Pointwise bound. Fix $1 \leq q \leq Q$, a unit a and $1 \leq z \leq x$. If $z < \sqrt{x}$, then Lemma 5.1 gives $|\psi(z; q, a) - z/\varphi(q)| \ll z \leq \sqrt{x}$. If $z \geq \sqrt{x}$ (so $z \geq 2$), Lemma 3.12 gives

$$\psi(z; q, a) - \frac{z}{\varphi(q)} = \Delta_\Lambda(z; q, a) + \frac{1}{\varphi(q)} \left(\sum_{\substack{n \leq z \\ (n, q)=1}} \Lambda(n) - z \right),$$

and by Theorem 4.4 with $B = A + 2$, Lemma 4.5, $\log q \leq L$ and $\log z \leq L$, the bracket is $\ll_A R := x/L^{A+2} + L^2$ in absolute value. Hence, using Lemma 3.18,

$$E(x; q) \ll_A \max_{\sqrt{x} \leq z \leq x} |\Delta_\Lambda(z; q, a)| + \frac{R}{\varphi(q)} + \sqrt{x}.$$

Summation. Sum over $q \leq Q$: the first terms are bounded by Theorem 10.2, i.e. by $\ll_A x/L^A$; by Theorem 5.6 the second terms sum to $\ll RL$; and the third terms sum to $\ll Q\sqrt{x} \leq x/L^{A+3} \leq x/L^A$. Finally

$$RL = \frac{x}{L^{A+1}} + L^3 \leq \frac{x}{L^A} + (A+3)! \frac{x}{L^A},$$

where the last step uses $L^{A+3}/(A+3)! \leq e^L = x$. $\square$

Theorem 10.6 (The Bombieri–Vinogradov theorem). *For every $A \in \mathbb{N}$, all real $x \geq 2$ and all real $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \sup_{1 \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}.$$

In Lean the left-hand side is a sum of the extended-real quantities $E(x; q)$ of Definition 10.3; the real-valued form is Corollary 10.7.

Proof. If $x \leq e^{32}$, apply Lemma 10.4.

If $x > e^{32}$, then $L := \log x \geq 32$. Put $U = V := e^{\sqrt{L}}$. Then (x, U, V) is a parameter datum (Definition 3.13): $x \geq 2$, $U, V \geq e^{\sqrt{L}}$ trivially, and $UV = e^{2\sqrt{L}} \leq e^{L/2} = \sqrt{x}$ because $2\sqrt{L} \leq L/2$ for $L \geq 16$. Now Lemma 10.5 gives the bound. $\square$

Corollary 10.7 (Bombieri–Vinogradov, real form). *For every $A \in \mathbb{N}$, all real $x \geq 2$ and $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \sup_{1 \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A},$$

where now the suprema are taken in $\mathbb{R}$.

Proof. Each $E(x; q)$ with $q \geq 1$ is finite (Definition 10.3), so the extended-real inequality of Theorem 10.6 may be transported to $\mathbb{R}$ by taking real parts (`ENNReal.toReal`) on both sides. $\square$